

Prague University of Economics and Business

Faculty of Informatics and Statistics



# **Comparison of feature importance methods for microbiological data**

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Author: Marek Havelka

Supervisor: RNDr. Ing. Petr Máša, Ph.D.

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## **Abstract**

This bachelor thesis focuses on the comparison of feature importance methods on microbiological data. The aim of the work is to compare the accuracy of the prediction model on a given dataset when manipulating the data according to the selected feature importance methods. Specifically, the SHAP and ARA methods are considered. The theoretical part of the thesis contains definitions of terms and description of the two selected feature importance methods. The methodological part describes how the research of the given issue was carried out, contains a general description of the methods used and their practical application in the practical part of the work. The practical part is implemented in the form of data processing using the Python language in Jupyter Notebook. These tools are used to measure the accuracy of the Catboost prediction model in adjusting the dataset according to importance using the SHAP and ARA methods, as well as factor analysis or Principal Component Analysis.

## **Keywords**

Feature importance method, data, prediction model, dataset

## Table of contents

|                                                                                                                   |    |
|-------------------------------------------------------------------------------------------------------------------|----|
| Introduction .....                                                                                                | 6  |
| 1 Theoretical section .....                                                                                       | 7  |
| 1.1 Feature .....                                                                                                 | 7  |
| 1.2 Black-box model .....                                                                                         | 7  |
| 1.3 Interpretable machine learning .....                                                                          | 7  |
| 1.3.1 Feature importance .....                                                                                    | 8  |
| 1.3.2 Feature Attribution .....                                                                                   | 8  |
| 1.3.3 Interpretable machine learning methods .....                                                                | 8  |
| 1.3.4 Kaiser's Rule .....                                                                                         | 9  |
| 2 Methodological section .....                                                                                    | 10 |
| 2.1 SHAP .....                                                                                                    | 10 |
| 2.2 ARA .....                                                                                                     | 11 |
| 2.3 Permutation feature importance .....                                                                          | 12 |
| 2.4 PCA .....                                                                                                     | 12 |
| 2.5 Factor analysis .....                                                                                         | 14 |
| 2.6 KMO and Bartlett's test .....                                                                                 | 16 |
| 2.6.1 KMO test .....                                                                                              | 16 |
| 2.6.2 Bartlett's test .....                                                                                       | 16 |
| 2.6.3 Effect of correlation on the importance of attributes .....                                                 | 16 |
| 2.7 AOPC .....                                                                                                    | 17 |
| 3 Practical section .....                                                                                         | 19 |
| 3.1 Description of dataset .....                                                                                  | 19 |
| 3.2 Dataset processing .....                                                                                      | 19 |
| 3.2.1 Initial processing .....                                                                                    | 19 |
| 3.2.2 Shap and ARA feature importance .....                                                                       | 20 |
| 3.2.3 Initial training and measurement of model prediction accuracy .....                                         | 20 |
| 3.3 Comparison of model accuracy with SHAP and ARA adjusted datasets by removing attributes without returns ..... | 21 |
| 3.3.1 Measurement of the accuracy of the model during successive attribute removal for SHAP .....                 | 22 |
| 3.3.2 Output and visualization of model accuracy measurements during attribute removal per SHAP .....             | 22 |
| 3.3.3 Calculation of the area under the curve for SHAP .....                                                      | 23 |

|                                                                                                              |    |
|--------------------------------------------------------------------------------------------------------------|----|
| 3.3.4 measurement of the accuracy of the model during successive attribute removal for ARA.....              | 23 |
| 3.3.5 Output and visualization of model accuracy measurements during attribute removal per ARA.....          | 23 |
| 3.3.6 Calculation of the area under the curve for ARA .....                                                  | 24 |
| 3.3.7 Summary of measurement results for SHAP and ARA methods for attribute removal without returns .....    | 25 |
| 3.4 Comparison of model accuracy with SHAP and ARA adjusted datasets by incremental adding attributes.....   | 26 |
| 3.4.1 Incremental addition of attributes to the dataset.....                                                 | 26 |
| 3.4.2 Incremental addition of attributes to the dataset output and visualisation .....                       | 27 |
| 3.4.3 Summary of method comparisons for incremental attribute addition.....                                  | 29 |
| 3.5 Summary of the comparison of SHAP and ARA methods using dataset adjustment                               | 30 |
| 3.6 Dataset modification using Factor and Principal Component Analysis.....                                  | 31 |
| 3.6.1 Dataset preparation for factor analysis .....                                                          | 31 |
| 3.6.2 Factor analysis.....                                                                                   | 34 |
| 3.6.3 Principal Component Analysis.....                                                                      | 35 |
| 3.6.4 Performing principal component analysis.....                                                           | 36 |
| 3.6.5 Attribute removal based on component membership and feature importance .                               | 37 |
| 3.6.6 Performing SHAP, ARA and permutation feature importance method on the output of the PCA analysis ..... | 39 |
| 3.6.7 ARA method .....                                                                                       | 43 |
| 3.7 Summary of the results of PCA and factor analysis.....                                                   | 45 |
| Conclusion .....                                                                                             | 48 |
| List of graphs .....                                                                                         | 52 |
| List of images.....                                                                                          | 53 |
| List of tables.....                                                                                          | 54 |
| Literature used .....                                                                                        | 55 |

# Introduction

Given the increasing role of artificial intelligence and the need to process ever-larger volumes of data, it is important to focus on accurate processing, understanding and interpretation. Example of this is biological or microbiological data. Their processing and interpretation are complex, and in recent years there has been a growing demand for the ability to use them effectively, and therefore for the ability to work with them efficiently.

I chose a specific biological dataset to show that methods which explain the importance of individual attributes in datasets are crucial for proper understanding and working with the data. It is important to understand how each method works and what its limitations are. In particular, there may be problems with datasets which contain highly correlated attributes.

The aim of this thesis is to compare the performance of the prediction model depending on the methods SHAP and ARA which explain the effect of the attributes on the prediction model results. The research questions of this thesis are:

- Main research question: How does the feature importance assesment of the SHAP and ARA methods differ for the CatBoots black-box model, and which method provides better results for model interpretation?
- First secondary research question: Are here any factors that might negatively influence these results?
- Second secondary research question: How will the results of the methods change if factors negatively affecting results are removed?

This paper is divided into theoretical, methodological and practical sections. The theoretical section deals with the definitions used in the analysis of the methods. Specifically, this includes definitions of feature importance, interpretable machine learning or prediction model. The theoretical part therefore includes a search of available sources.

The methodological section describes the chosen approach for comparing methods. This section includes the method of comparison and measurement. Specifically, it involves measuring the accuracy of the prediction model when fitting the dataset according to SHAP or ARA, calculating the area under the curve of accuracy for each model and comparing. It then goes on to describe factor analysis and PCA.

The practical section includes the application of the above-mentioned methods on a specific microbiological dataset. In the practical part, python code in Jupyter Notebook was used. The code and individual parts of the analysis are available from: [https://1drv.ms/f/c/dee8f4d7238ba781/EjcZeaHXtjVBp\\_2ca\\_YLdaEBenoj13xDSzOCA2QUHIRB.Jw?e=ORExbg](https://1drv.ms/f/c/dee8f4d7238ba781/EjcZeaHXtjVBp_2ca_YLdaEBenoj13xDSzOCA2QUHIRB.Jw?e=ORExbg)

# 1 Theoretical section

This section of the thesis deals with the definition of prediction model, black-box model, feature, interpretable machine learning, feature attribution method and feature importance method in general within machine learning and prediction models. The sources used in this section are professional articles and scientific journals dealing with selected issues. The obtained data are processed using the literature search method.

## 1.1 Feature

Before we can define the feature importance method, we have to define the feature itself. The feature can be defined as a property of a recorded dataset. This property is measurable and it can also be called a variable or attribute of a dataset. Every feature has a certain level of bearing on a model's use case and this bearing we can call feature importance. (Domino, 2025)

As a feature, we can call either individual variables or variables derived from individual variables or combined attributes. These combined attributes are derived from underlying data elements. (Domino, 2025)

## 1.2 Black-box model

Due to the fact, that the practical section of the thesis deals with the Catboost model, which is a type of Black-box model it is needed to define what a black-box model is and how it is connected to Interpretable machine learning.

Black-box models are models which lack transparency. It means that people can have trouble understanding how a model arrived at its results. People can operate only either with a dataset or the output of a model so it cannot be directly explained. Despite of that this type of model provides very accurate predictions and the predictions are high quality. (Adadi & Berrada, 2018).

Insides of the model are however essential for practitioners. With an understanding of how any model works, they can ensure, that a model works correctly, for example by setting model parameters. They can also ensure that a model respects social, legal or ethical aspects. (Matějka, 2024)

## 1.3 Interpretable machine learning

In the past decade, predictive models have been becoming more and more complex, because of our need to use them for large and complex datasets. This led to concern, that people would not be able to understand the reasoning of these models.

Because of these concerns, grow out the field of interpretable machine learning. The goal of this field is to empower and build trust in models, perform model debugging and enable to people understand how models work. The field of interpretable machine learning is focused on the development and evaluation of quantifiable technical objectives, however as Valerie Chen et al. (2022) mention, it is important to measure the effectiveness of these methods on concrete use cases, as it is in the practical section of this thesis. (Valerie Chen et al., 2022)

A specific case of how interpretable machine learning can provide insight into a trained model is by describing the behaviour of the model using feature relationships. More specifically by explaining feature attribution or feature importance. (Valerie Chen et al., 2022)

### **1.3.1 Feature importance**

Feature attribution and feature importance are concepts, that provide insight into a model by explaining the contribution of individual features, however, they differ in approach. Feature importance measures the overall contribution of a feature, typically by using permutation feature importance (for example SHAP) and it assesses the impact of modifying or removing a feature on a model's predictions and provides a score or ranking which reflects the importance of a feature. Then we can name the most relevant features in a dataset. (Massed Compute, 2024)

### **1.3.2 Feature Attribution**

Feature attribution provides a more complex explanation of how a feature is important for a model. It seeks answers to questions like: *Which words in the sentence led to the model's classification of the text as spam?* Methods based on feature attribution calculate the contribution of a feature by analysing the gradients of the model's loss function with respect to the input features. (Massed Compute, 2024)

Feature attribution method is meant to explain a prediction of black-box models such as neural networks. There are several types of these methods and due to their conceptual differences, these different methods provide ambiguous explanations. This allows for obtaining numerous different insights into the result of a black-box model, it also makes the decision of what method to adopt more difficult. (Mandler et al., 2024)

Overall we can conclude that feature attribution methods quantify for each feature its importance for a given model. This allows ranking features of a given dataset according to their importance or identifying these features as non-contributory for the prediction at all. (Mandler et al., 2024)

### **1.3.3 Interpretable machine learning methods**

Within interpretable machine learning are several types of methods which are used. These methods differ in whether they consider features of interest in isolation, an example of this



type is Permutation feature importance (PFI), or in relation to all remaining feature variables, for example, conditional feature importance (CFI). (König et al., 2021)

#### **1.3.4 Kaiser's Rule**

Kaiser's rule is related to the problem of selecting the ideal number of factors or components for factor or PCA analysis. It is only one of several variations that can be used to select the number of factors in the exploratory phase of factor analysis. An alternative approach is the scree test, a graphical method in which the choice of the number of factors is given by a fraction in the factor eigenvalue curve. (Budíková, Koutková, & Dan, 2014)

Kaiser's rule works with the assumption that if all variables are completely independent all eigenvalues of the solution will be unity. Given this, it is appropriate to choose as many factors as have eigenvalues greater than one. (Soukup, 2021)

## 2 Methodological section

The methodological part of this thesis describes the specific methods of comparison, observation or measurement used in the practical part. Specifically, this part deals with the description of SHAP and ARA methods. It also describes PCA and factor analysis, KMO and Bartlett's test and AOPC.

### 2.1 SHAP

SHAP(Shapley Additive exPlanations) is an approach to explain the output of a machine learning model by connecting optimal credit allocation with a local explanation. It is based on Shapley values using game theory to assign credit for each feature value to the model's prediction. (SHAP, 2018)

Conceptually, we can say that if we have a coalition which collaborates to produce a target value then SHAP is an approach to explain how each member of the coalition contributes to the target value.(KIE, 2021)

First of all, we need to describe the Shapley values themselves. Within game theory, Shapley values are a method of fairly distributing the gains of a game, so that each link contributing to the gains gets a fair share corresponding to its involvement. (Yadav, 2024)

To determine the contribution of an individual attribute, the coalition when the attribute is removed is compared with the coalition containing all attributes. This difference is referred to as the Marginal Contribution of a removed attribute to the coalition. And it expresses how much a given attribute contributed to a given coalition. The Shapley value for a given attribute is then the average of all possible combinations of coalitions where the attribute is removed. This is done for all attributes. (KIE, 2021)

This whole concept was used for the purpose of explaining the models. While at the conceptual level, it is about explaining the contribution of coalition members to coalition value for the purpose of prediction models, it is about explaining the contribution of individual attributes to the model output. (KIE, 2021)

The authors of the paper that deals with translating this concept for the purpose of interpreting prediction models are Scott M. Lundberg and Su-In Lee in their paper "A Unified Approach to Interpreting Model Predictions". (Lundberg & Lee, 2017)

As part of their work, they introduced Additive Feature Attribution Methods. This method was based on creating simplified models to explain more complex models such as neural networks. Quite specifically, these are linear functions of binary variables. (Lundberg & Lee, 2017)

To simplify, we can think of it this way. If there is a set of input  $z$  and a model  $f(z)$ , we can define a set of simplified variables  $z'$ . These simplified variables are typically a feature vector turned into a binary vector. (Lundberg & Lee, 2017)

It is then possible to define an exponential model  $g(z')$ . Importantly, however, if  $z$  is approximately equal to  $z'$ , then  $f(z)$  is approximately equal to  $g(z')$  and at the same time  $g(z')$  must be of the form:

$$g(z') = \phi_0 + \sum_{i=1}^M \phi_i z'_i$$

Formula adapted from Lundberg and Lee (2017).

Where  $z \in \{0, 1\}$  and  $M$  is the number of simplified input features, and  $\phi_i \in \mathbb{R}$ . (Lundberg & Lee, 2017)

As an attribution itself, we can define as  $\phi_i$ . (KIE, 2021)

There are 3 desirable properties of an additive feature method. It is local Accuracy, Missingness, and Consistency. First, of these properties – local accuracy is connected to the requirement, that if  $z = z'$  then  $g(z') = f(z)$ . In other words, if input and simplified input are at least roughly the same then we can say that the model and explanatory model will give roughly the same output. (KIE, 2021)

The second property is Missingness and it states, that if some feature is excluded then the contribution of the feature must be zero. So if simplified inputs represent the presence of a feature then the absence of a feature requires an absence of simplified input. (KIE, 2021 ; Lundberg & Lee, 2017)

$$x'_i = 0 \Rightarrow \phi_i = 0$$

Formula adapted from Lundberg and Lee (2017).

Consistency is the third property. This property says that if any simplified input's contribution increases by changing a model the input's attribution in the explanatory model should not decrease. (Lundberg & Lee, 2017)

Since the SHAP method is a post hoc method, i.e. a method that is performed after the model has been trained, there might be some doubts about the interpretability of the data based on these kind of method. (Ragodos, Wang, Feng, & Hu, 2024)

## 2.2 ARA

It is a method that is based on association rules to examine the effect of a dataset attribute on the target variable. This exploration takes place at several levels. At the beginning, the influence of the attribute itself is examined. If the importance of an attribute exceeds a

certain threshold, then its importance is examined at the next level in conjunction with other attributes. (Araxai, 2024).<sup>1</sup>

## 2.3 Permutation feature importance

Permutation feature importance is a model based on the technique of permutation relevance. It is a technique that measures the contribution of each element of a dataset in relation to the performance, i.e. accuracy, of a given model. This technique is suitable for non-linear or opaque estimators. The principle of this technique is to randomly mix the values for a given element. In other words, for each attribute, its data will be shuffled thereby breaking the relationship between the attribute and the target variable. By breaking the relationship between the target variable and the attribute, it is then possible to determine how important the attribute is to the accuracy of the model.(scikit-learn, 2025)

The great advantage of this method is that it can be applied to any model, it is also simple to implement and can be computed multiple times with different permutations.(scikit-learn, 2025)

It is important to note that attributes that may be of little importance for one model may be very important for another model. It is therefore desirable to calculate the predictive power of the model using cross-validation before permutations.(scikit-learn, 2025)

Mathematically, the technique can be described as follows. Let there be a model  $m$  and a dataset  $D$ . A reference score  $s$  is computed. For each attribute  $j$  of dataset  $D$ , at each iteration  $k$  in  $1...K$ , a column/attribute  $j$  of dataset  $D$  is randomly shuffled and a modified dataset  $D_{k,j}$  is generated. Compute the score  $S_{k,j}$  of model  $m$  on corrupted data  $D_{k,j}$ .(scikit-learn, 2025)

This method was used as a complementary method to SHAP and ARA in the analysis of the impact of independent attributes.

## 2.4 PCA

Principal component analysis is a tool for reducing variables and also helps to uncover hidden variables for better interpretation of data.( portal.matematickabiologie, 2025) The goal of PCA is to replace a large number of variables with a smaller number to reduce correlations in the data. A sufficiently strong correlation in the data is a prerequisite for using PCA.(Ambrožová, 2016)

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<sup>1</sup> At the time of writing, the description of this method is limited by the scope of the description in the official documentation of the method.

The principle on which PCA is based is that a smaller number of principal components are created from the original attributes of the dataset as a linear combination of the original attributes. The main characteristic of a component is then its variance.(Solnický, 2008)

In addition to the principal component, the concepts of eigenvector, eigenvalue and load are important. The eigenvector is the set of coefficients of the linear equation of the principal component. The eigenvalue indicates in absolute units what part of the variance is explained by the component. The loading is then the relationship between the original variable and each component. According to the loadings, it is possible to determine which component a given attribute falls most heavily into. (Ambrožová, 2016)

Klára Ambrožová in her presentation *Analýza hlavních component* describes the recommended procedure for PCA as follows.<sup>2</sup> Using graphs to determine if variables have different variability, then creating matrix graphs to identify correlations. Here it is worth noting that an alternative to matrix plots for identifying correlations may be to perform KMO and Bartlett test. The next step is to select the appropriate number of principal components using the Kaiser criterion, i.e., selecting components with eigenvalue greater than 1. Here, it is also possible to identify the ideal number of components by performing PCA with a random number of components and using visualization to find the elbow. This value is the optimal number of components.(Ahmed, 2025)

The next step is to perform the analysis itself and evaluate the results. In the author's case, the software was STATISTICA, but it is also possible to use the Python library sklearn.decomposition. (Ambrožová, 2016)

The analysis itself can be described mathematically as follows: If there is a dataset  $x$  with  $n$  rows and  $p$  attributes, then the covariance matrix that captures the linear relationships between the attributes of the dataset is calculated as  $x_c = x - \bar{x}$ , where  $x_c$  are centred data and  $\bar{x}$  is a mean vector of each feature. Then covariance matrix is:

$$\Sigma = \frac{1}{n-1} x_c^T x_c$$

Formula adapted from Sharma (2024).

Where  $x_c^T$  is the transpose of the centered matrix  $X_c$ . (Sharma,2024)

The covariance matrix allows eigenvalues decomposition. This decomposition involves finding eigenvalues ( $\lambda$ ) and eigenvectors ( $v$ ). The decomposition is defined as:

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<sup>2</sup> It should be noted that this presentation was created for the needs of the course Statistical Methods II, 8 exercises at Masaryk University. Thus, the mentioned procedure may be subject to didactic or technical requirements for teaching purposes.

$$\Sigma = v \Lambda v^T$$

Formula adapted from Sharma (2024).

Here  $\Lambda$  is a diagonal matrix with eigenvalues and  $V$  is the matrix of eigenvectors. The principal components are then eigenvectors ordered by the size of their eigenvalues. (Sharma,2024)

To reduce the dimensionality or correlation of dataset attributes, we need to select  $k$  eigenvectors corresponding to the  $k$  largest eigenvalues and transform the centred  $X_c$  data onto these principal components:

$$x = x_c v_k$$

Formula adapted from Sharma (2024).

Where  $V_k$  is a matrix composed of the top  $k$  eigenvectors. (Sharma, 2024)

## 2.5 Factor analysis

Factor analysis is a statistical method, which is used to reduce a large number of attributes or variables into a smaller number of factors. The prerequisites necessary for the use of factor analysis for a particular dataset are: linearity, absence of multicollinearity, inclusion of relevant variables and sufficient correlation between attributes and factors. (Statistics Solutions, n.d.)

In contrast to PCA, factor analysis assumes that correlations between variables are not the result of correlations between attributes, but are the result of common hidden factors. At the same time, factor analysis attempts to explain correlations using fewer common factors, whereas PCA only explains the variance of the original variables. (Masarykova Univerzita, n.d.)

Factor analysis works similarly to PCA in the sense that it uses a covariance matrix to find the main factor that explains the largest part of the variance of the data matrix, and other factors that are unweighted and uncorrelated are created to gradually exhaust the maximum of the total variance. (Masarykova Univerzita, n.d.)

While the calculation of PCA is straightforward in the case of factor analysis, the procedure is more complex and there are more procedures for it. (Masarykova Univerzita, n.d.)

It is also important to note that in certain situations, factors may be correlated to some extent, as opposed to PCA components. Factor correlation can occur when factors are extracted from highly correlated variables and at the same time rotation is used to describe the variables as simply as possible by the factors. (Masarykova Univerzita, n.d.)

Mathematically, the influence of the factors can be explained as follows. If  $x^T = (x_1 + x_2 + x_3 \dots x_p)^T$  is an object described by  $p$  variables, then the general model of factor analysis assumes that there are  $m$  factors  $F_1, F_2 \dots F_m$ , which are less than  $p$ . The given object can be described by a linear combination of factors as follows:

$$x_1 = l_{11}F_1 + l_{12}F_2 + \dots l_{1m}F_m + e_1$$

$$x_2 = l_{21}F_1 + l_{22}F_2 + \dots l_{2m}F_m + e_2$$

$$x_p = l_{p1}F_1 + l_{p2}F_2 + \dots l_{pm}F_m + e_p$$

Formula adapted from Masarykova Univerzita (n.d.).

$F_1, F_2 \dots F_m$  are therefore factors that create correlations between  $p$  variables. These factors have zero mean and unit variance. The error components  $e_1, e_2 \dots e_m$  are specific factors, that contribute to the variance of the variables. (Masarykova Univerzita, n.d.)

The  $l_{ik}$  coefficients are factor weights or loadings. These loadings can be interpreted as the contribution of the  $j$ -th factor to the  $i$ -th variable and thus represent the correlation between the original and new variables - factors. (Masarykova Univerzita, n.d.)

Such a factor model can be rewritten in matrix form  $X = FL + E$ . Where  $X$  is the original data matrix,  $L$  is the matrix of factor loadings,  $F$  is the matrix of common factors and  $E$  is the error matrix. (Masarykova Univerzita, n.d.)

The covariance matrix  $S$  can be expressed as  $S = LL^T + I^2$ . This relationship can be explained by saying that  $LL^T$  is the part of the variability of the data explained by factors, while  $I^2$  is the uniqueness matrix and is the part of the variance that is not explained by factors. In the context of this mathematical explanation, it should be mentioned that in the case of PCA, the matrix is a singularity  $I^2$  is zero and thus that all PCA components explain the total variance of the dataset. (Masarykova Univerzita, n.d.)

In the practical application of factor analysis, the determination of the number of factors, as in the case of PCA, is subject to expert knowledge, the use of a scatter plot or the use of the Kaiser rule. (Masarykova Univerzita, n.d.)

For factor analysis, the interpretation of the results is different from PCA, which is due to the rotation of the factors. Factor rotation aids interpretation by ensuring that for each variable there is a strong association with only one factor, while for the other factors, the association with the variable is minimised. (Masarykova Univerzita, n.d.)

For factor analysis, the interpretation of the results is different from PCA, which is due to the rotation of the factors. Factor rotation aids interpretation by ensuring that for each variable there is a strong association with only one factor, while for the other factors the association with the variable is minimised. There are several ways of factor rotation, which

are classified according to whether the independence of the factors is maintained or whether the factors can be correlated to some extent. <sup>3</sup>(Masarykova Univerzita, n.d.)

## **2.6 KMO and Bartlett´s test**

Kaiser-Meyer-Olkin (KMO) and Bartlett´s Test are commonly used statistical tests used before factor or PCA analysis to get the suitability of data for these analyses. Their aim is to assess the degree of data correlations. (DATAPOTT Analytics, n.d.)

### **2.6.1 KMO test**

KMO test measures sampling adequacy for each variable in the model and for the complete model. The test examines the patterns of correlations between variables and compares the magnitude of the correlation coefficients with the magnitude of the partial correlation coefficients. (Statistic How To, n.d.)

### **2.6.2 Bartlett´s test**

This test assesses if there are significant relationships between the variables in a dataset. More specifically it tests the hypothesis if the correlation matrix of the variables is an identity matrix (values which are off-diagonal are zero). The null hypothesis is that the variables are uncorrelated and so the correlation matrix is an identity matrix. (DATAPOTT Analytics, n.d.)

### **2.6.3 Effect of correlation on the importance of attributes**

It should be mentioned here that correlation can have a negative effect when interpreting the importance of individual attributes. It can also be said that correlation affects the results of feature importance methods. There are studies that confirm that at different levels of correlation, the importance of individual attributes can be variable for different models. (Hooker, Mentch, & Zhou, 2019)

Thus, in the presence of correlation or even multicollinearity, changing one attribute has little effect on the model because the same information from one attribute can be obtained from correlated attributes. Importantly, the impact on model performance varies depending on the underlying implementation. However, one general best practice is to perform clustering and work with only one attribute from each cluster based on the chosen threshold. (Scikit-learn, n.d.)

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<sup>3</sup> If we consider PCA as a kind of factor analysis, then PCA is always orthogonal and there are no correlations between the components.



## 2.7 AOPC

Area over the perturbation curve (AOPC) is one of the methods/metrics for examining the quality of prediction models. It should be noted that the evaluation of the quality of prediction model results is still an active area of research and many authors, e.g. Cristian Cosci in his paper *Area Over Perturbation Curve using Most Relevant feature (AOPC)* mention that there is no standardised method for objective evaluation of prediction models and different defined metrics or procedures need to be related and adapted to the specific task. (Cosci, 2023)

To support the claim that methods for examining the quality of the outputs of prediction models must be tailored to the specific task, one can present the work of Joakim Edin and colleagues who, in their paper *Normalized AOPC: Fixing Misleading Faithfulness Metrics for Feature Attribution Explainability*, argue that AOPC, as a frequently used method, can lead to incorrect conclusions and present a modified version of AOPC, i.e., normalised AOPC. Their study pointed out the potential biases in using the standard AOPC and also the problematic interpretation of scores without knowing the upper and lower bounds of the model. It should be noted here again that the method for evaluating model output must be individualized to the needs of the problem at hand, and the standard AOPC is still a widely used method for evaluating model output. Thus, the standard AOPC is applied in the practical part of this paper. (Edin et al., 2024)

AOPC measures the change in the output of a given model as attributes are permuted over time. The permutation itself may involve removing, inserting, or replacing an element with another predefined element. (Edin et al., 2024)

Mathematically it can be expressed as follows:

$$\text{AOPC}(f, x, r) = \frac{1}{N} \sum_{i=1}^N f(x) - f(p(x, r_{1:i}))$$

Formula adapted from Edin et al (2024).

Where  $f$  is a model,  $x$  is a vector with  $N$  attributes,  $r$  is the permutation order ( $\{1 \dots N\}$ ) of the input features,  $p$  is a permutation, i.e. a function that inserts, replaces or removes features, which are in  $r_{1:i}$ . (Edin et al., 2024)

The permutation procedure itself may be different for different specific models and assignments, but the evaluation method is important. For the purposes of this work, the procedure for AOPC was that the original dataset was modified in various ways and changes in model accuracy were monitored. The actual evaluation was based on the measured model outputs and their visualization. This took the form of graphs, which included either changes in model accuracy or their absolute amount. In addition to the accuracy itself, the area under the accuracy curve was also calculated in some cases using numerical integration. However, it is important to note that the calculation of the area of integration does not help the interpretation of the model because it does not reveal important attributes. However, it can

be used to compare the outputs of two methods for identifying important attributes, e.g. SHAP and ARA, where it can be assumed that when measuring the absolute accuracy of the model under permutation, a method that incorrectly identifies important attributes would have a significantly higher area under the curve than a method that correctly identifies important attributes. When analysing the outputs using visualisation, i.e. graphs, the better the method that explains the importance of attributes, the faster the model accuracy curve decreases. (Cosci, 2023)

## 3 Practical section

### 3.1 Description of dataset

In the practical section, The author has worked with a dataset of microbiological data. The dataset contains 1684 columns/attributes and 25739 records. The target variable is called medium. This target is predicted based on binary attributes. Individual records represent identifiers from the NCBI Taxonomy database. Generally, these records are systematic classifications of organisms. The number in classification represents a particular organism or taxonomic group. (National Center for Biotechnology Information, n.d.)

Attributes of dataset are mostly from the ChEBI database – Chemical Entities of Biological Interest. This database provides information about chemical compounds with relevance to biology and biochemistry. An identifier refers to a specific molecule, ion, metabolite or other chemical entity. (European Bioinformatics Institute, n.d.)

Another type of attribute is trophic strategies. These attributes/strategies define the ways in which organisms obtain energy and carbon for their metabolism and growth. (Masarykova univerzita, 2015)

The binary nature of the attributes describes either whether the organism is associated with the chemical entity which is represented by 1 or whether it is not associated with the entity. This applies to those attributes that are marked as CHEBI and are therefore records from the ChEBI database. In the second case, attributes describe whether the attributes use a specific trophic strategy. If a given attribute is 1, then it means that the organism uses the trophic strategy, if the attribute is 0, then it does not. This applies to those attributes that are marked as trophic\_type.

### 3.2 Dataset processing

The initial part of the dataset processing is to modify it to allow training and use of the catboost predictive model. The initial training of the model was crucial to obtain model accuracy. This model accuracy later acted as a benchmark for further processing of the dataset and models re-trained on it. In terms of the research questions, the comparisons can be used to answer how the feature importation assessment differ for different methods of a given model, and secondarily to see if there are effects that bias the results.

#### 3.2.1 Initial processing

In the processing of the dataset, in addition to loading the dataset itself, the initial step was standard to split the dataset into two parts. The first part (X) contains the part of the dataset on the basis of which the prediction model later predicted the target variable, which was

separated separately into the second part(y). In the next step, both parts were split into test, training and validation kits. Furthermore, the target variable was converted into binary form within the initial processing.

### 3.2.2 Shap and ARA feature importance

Not only in the initial part of the thesis has been worked with attributes sorted by feature importance according to SHAP and ARA methods. These lists were created directly on the original dataset. The list for ARA contained 98 of the most important features of the dataset. On the other hand, the list for SHAP had 47 of the most important features. This is important for further processing and especially for comparing because in comparison it was necessary to compare SHAP with only the first 47 attributes for the ARA method.

The author has been provided with sorted attributes for SHAP and ARA in .xlsx format however, for further processing, it was more practical to convert these data from .xlsx format to NumPy array.

### 3.2.3 Initial training and measurement of model prediction accuracy

Before any comparison of feature importance methods, it was necessary to initially train the model and measure its accuracy. Further in the thesis, I compared the accuracies of models trained on modified datasets and compared these accuracies with the accuracy of this initial model.

Image 1 - Model with set parameters used in initial model training as well as in training on modified datasets for later comparison (own processing)

```
# run the model once to get the overall accuracy
%%time
model = CatBoostClassifier(random_seed=RANDOM_SEED,
                           iterations=100000,
                           loss_function="Logloss",
                           learning_rate=0.05, #0.001
                           depth=4,
                           l2_leaf_reg=4,
                           bagging_temperature=1,
                           random_strength=6,
                           verbose=100)

model.fit(train_data,
          eval_set=val_data,
          early_stopping_rounds=50
          ), plot=True)
```

Image 2 - This is the accuracy of the initial model. This accuracy was later used as a benchmark (own processing)

```
Accuracy: 0.9548842683405256
Balanced Accuracy: 0.8708789660926911

Classification Report:
              precision    recall  f1-score   support

     0       0.96       0.99       0.97       2212
     1       0.89       0.76       0.82        337

   accuracy          0.95       2549
  macro avg       0.92       0.87       0.90       2549
 weighted avg       0.95       0.95       0.95       2549

AUC-ROC: 0.9779326146564999
```

### 3.3 Comparison of model accuracy with SHAP and ARA adjusted datasets by removing attributes without returns.

This subsection describes the process of dataset adjustment, subsequent model training and model accuracy measurement as a way to compare SHAP and ARA methods.

In the first part of comparing the results of the SHAP and ARA methods, was measured the decrease in model accuracy when attributes were removed in sequence, separately for the SHAP and ARA methods.

This process aims to track the decrease in model accuracy and works with the assumption that if a given method correctly evaluates an attribute as important for feature importance, then a decrease in accuracy will occur. Moreover, if the method ranks the attributes by importance correctly, then the accuracy of the model will systematically decrease.

This procedure will also allow a comparison of which method is better at finding feature importance for each attribute.

Here can be stated that if one of the methods would have better results than the other method, but the decrease in accuracy would be small, then neither method can be used to interpret the behaviour of the black-box model since removing the attributes will not affect the performance of the model.

### 3.3.1 Measurement of the accuracy of the model during successive attribute removal for SHAP

This procedure was first applied for the SHAP method. The measurement procedure was as follows. An array to store the accuracy of each model in turn was created. Next, it was created a for loop that traversed the NumPy array with the attributes sorted according to the SHAP method. At each iteration of the loop, the attribute was removed from the dataset, the model was trained, and the model accuracy was measured.

It is important to mention here that there were no returns of discarded attributes. In other words, after removing an attribute from the dataset, the attribute was not returned to the dataset in the next iteration, but another attribute was removed from the list.

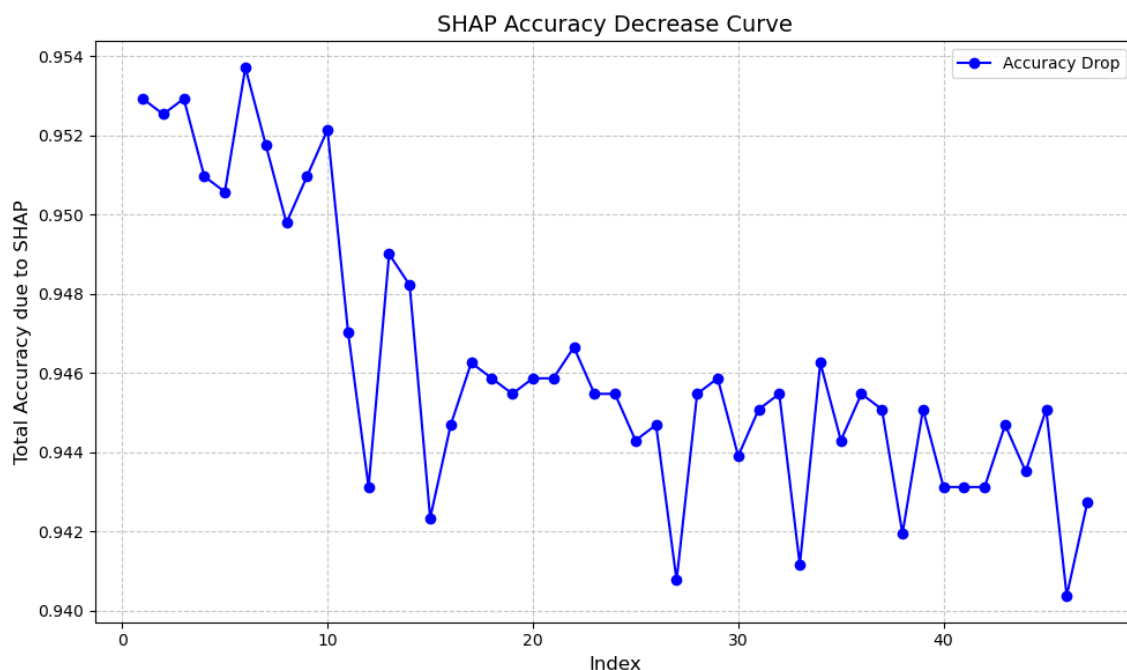
This procedure involved 47 iterations of the loop for 47 attributes according to the SHAP method.

### 3.3.2 Output and visualization of model accuracy measurements during attribute removal per SHAP

The output of the above procedure was the accuracies of the individual models. The accuracies that were stored in the array had to be visualized for the purpose of AOPC analysis.

In the graph below you can see the output of the above-mentioned procedure and therefore the decrease in model accuracy when attributes are removed.

Graph 1 - Decrease in model accuracy for the SHAP method without returning attributes (own processing)



The graph shows the accuracies of the models for a number of sampled attributes removed ranked by importance using the SHAP method.

Already in this first graph, it can be noticed that the decrease in model accuracies is not gradual although it is noticeable, that there are some differences in the accuracy of the model at the beginning and at the end of the procedure. One would expect that if the output of the SHAP method was completely correct or was performed on a different dataset, then the largest drop would occur when the first attributes were removed and the drop in accuracy would gradually decrease when less important attributes were removed. Here, however, there is not a gradual decrease but in certain cases, there is even an increase in accuracy at certain points.

We can assume that the uneven decrease in accuracy is due to the robustness of the model and the high correlation of dataset attributes.

### **3.3.3 Calculation of the area under the curve for SHAP**

The next step was to calculate the area under the curve of accuracies. This was made by numerical integration by trapezoid method using `np.trapz` function from NumPy Library.

The logic of the calculation is that for each point in the graph, a trapezoid of height equal to the accuracy curve is created. And then the individual trapezoids are added together.

The area under the curve for non-return accuracy for the SHAP method is approximately equal to 43.52.

The calculation of the area under the curve, as well as the curve accuracy itself, is a central part of the AOPC method.

### **3.3.4 measurement of the accuracy of the model during successive attribute removal for ARA**

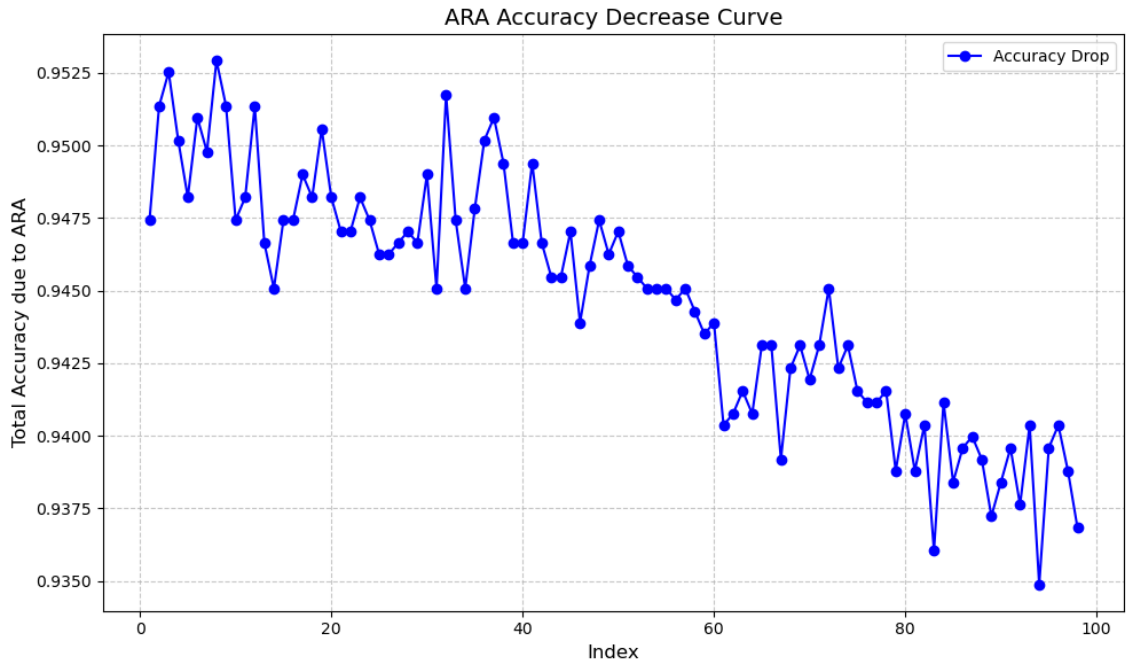
For the ARA method, the procedure was similar to the SHAP method. An array was created to store the accuracies. A loop was then created in which the attributes were removed in each iteration as they were ordered by the ARA method without returning the attributes.

This procedure involved 98 iterations of the loop for 98 attributes according to the ARA method

### **3.3.5 Output and visualization of model accuracy measurements during attribute removal per ARA**

The output of the procedure was the model accuracies for attribute removal ranked according to the ARA method. These individual accuracies were visualized as a graph once more.

Graph 2 - Decrease in model accuracy for the ARA method without returning attributes (own processing)



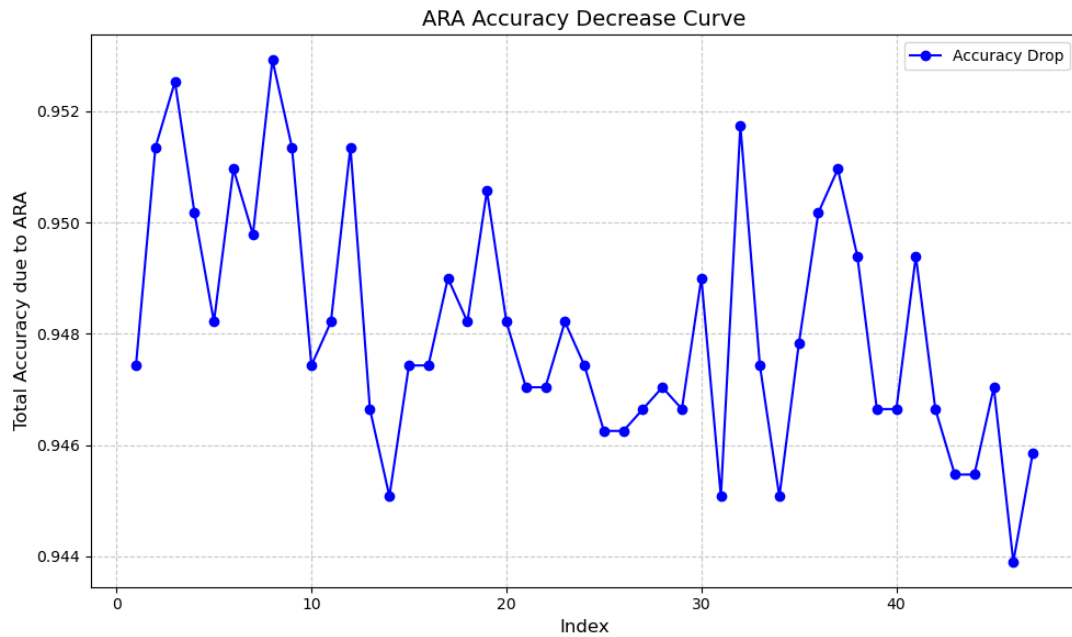
Again, it can be noticed that although there is a gradual decrease in the accuracy of the models, the decline is not straightforward. Repeatedly, there are increases at some points. And even in this case, the model trained at the beginning of the process, i.e. with the full dataset, is not much more accurate than the model trained at the end, i.e. after removing 98 attributes.

### 3.3.6 Calculation of the area under the curve for ARA

The output of the process had to be modified for the purpose of the calculation or the possibility of comparing the area under the curve for the ARA method. Due to the fact that there were only 47 attributes for the SHAP method, it was only necessary to work with the first 47 attributes for the ARA method.



Graph 3 - Decrease in model accuracy for the ARA method without returning attributes for the first 47 attributes (own processing)



The calculation of the area under the curve for the ARA method was the same as for the SHAP method and came out to approximately 43.61.

### 3.3.7 Summary of measurement results for SHAP and ARA methods for attribute removal without returns

From the accuracy measurements for both methods without attribute returns, it can be observed that there is a decrease in accuracy in both cases. However, in both cases, the decrease is not gradual, with increases in accuracy at some points. Also, in both cases, the accuracies of the models before and after the removal of all attributes ranked by importance according to both methods show only a small decrease, namely in tenths of percent or at most units of percent.

Based on the measurements described in this chapter, it can be concluded that neither method in this measurement performed significantly better than the other method. For both methods, there was only a small decrease between the first and last attribute. Thus, both methods have similar results and neither method can be used to interpret the model.

Table 1 - results of the SHAP/ARA sequential attribute removal procedure (own processing)

| method | Accuracy on first attribute removed | Accuracy at last attribute removed (47. Attribute) | Area under the curve |
|--------|-------------------------------------|----------------------------------------------------|----------------------|
| SHAP   | 0.95292271                          | 0.94272264                                         | 43.5221              |
| ARA    | 0.94743036                          | 0.94586112                                         | 43.6139              |

### 3.4 Comparison of model accuracy with SHAP and ARA adjusted datasets by incremental adding attributes.

It can be assumed that the results of the investigation of the effect of removing attributes from the dataset without returning them may be affected by the correlation of attributes, the next step was to create an empty dataset to which attributes were successively added in order of importance using the SHAP and ARA methods.

If it was true that the attributes of the dataset were strongly correlated, then it could be expected that adding attributes to the dataset sequentially would, despite their importance, increase the accuracy of the model slightly. At the same time, there would not be a steady increase in accuracy, but accuracy could increase by leaps and bounds at random points that correlate with other points.

This procedure allows us to re-examine the results of both methods, i.e. feature importance for individual attributes for the SHAP or ARA method, and possibly assess whether the model can be interpreted based on them. In contrast to the previous procedure, it is also possible to assess negative influences affecting the results of both methods. Specifically, the correlation between attributes. The question of whether correlation is present and what is its effect on the results of the methods arises from the previous procedure

#### 3.4.1 Incremental addition of attributes to the dataset

The procedure for training the model on the dataset to which attributes were added according to importance was the same for both methods. First, a new dataframe was created. This dataframe contained only the target variable that the model was about to predict. The dataset had the same indexing as the original dataset, so that individual attributes from the original dataset could be added and were correctly linked to the target variable.

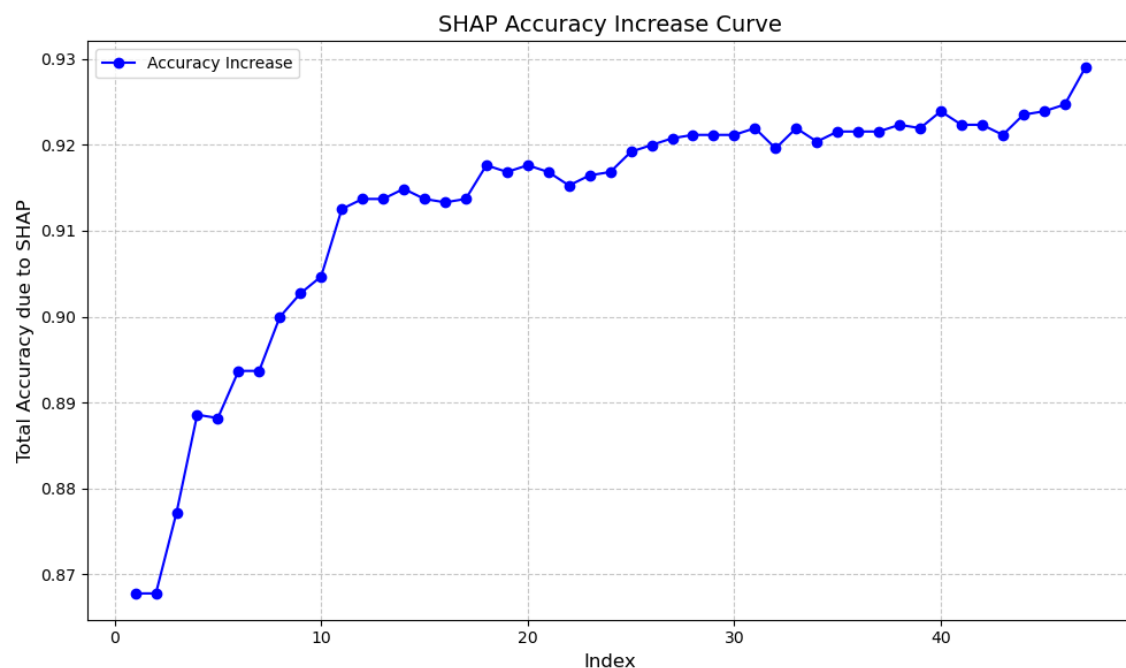
Then a loop was created within which an attribute was added to the dataset, according to the importance of the method. Thus, at the beginning the most important attribute for the

chosen method was selected and added to the dataset, then the model was trained and its accuracy was determined, which was stored in the prepared array. In the next iteration, the second most important attribute was added, the model was trained again and the accuracy was stored again. This process was repeated for all the attributes ordered by importance for the method.

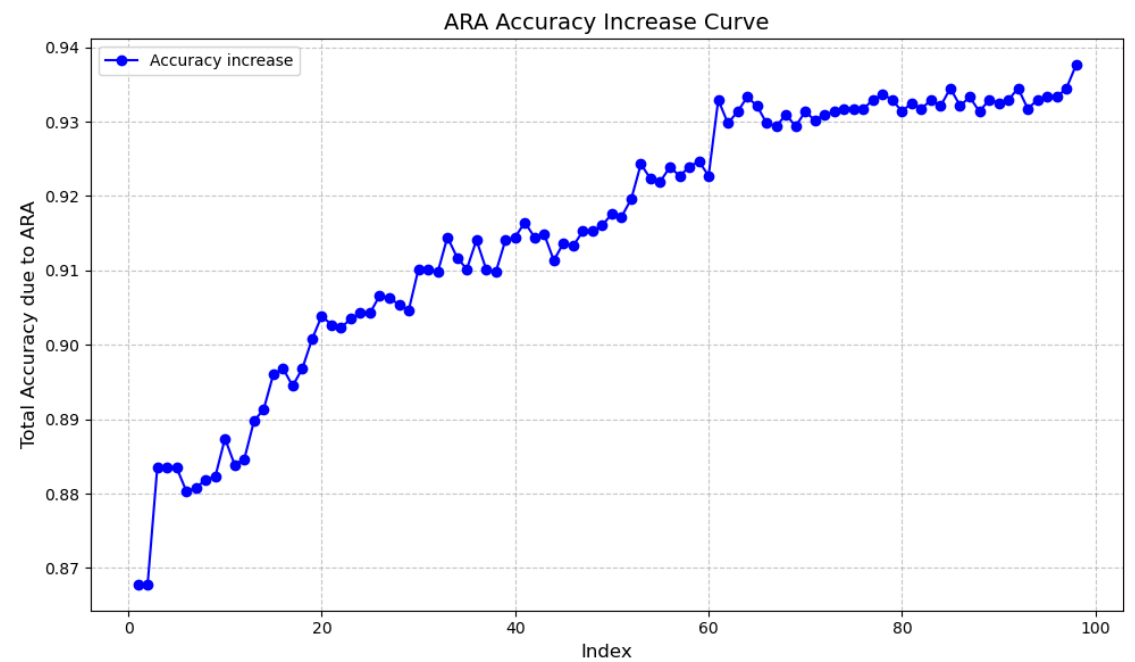
### 3.4.2 Incremental addition of attributes to the dataset output and visualisation

The output of the above description was the accuracies of the individual models. These accuracies have been visualised for both methods, i.e. SHAP and ARA, in the form of a graph on which the evolution of the model accuracies can be observed. For a given curve, the area under the curve was calculated using Numerical Integration with the trapezoidal rule to capture the accuracy of the models.

Graph 4 - Increase in model accuracy for the SHAP method as attributes are incrementally added to the dataset (own processing)

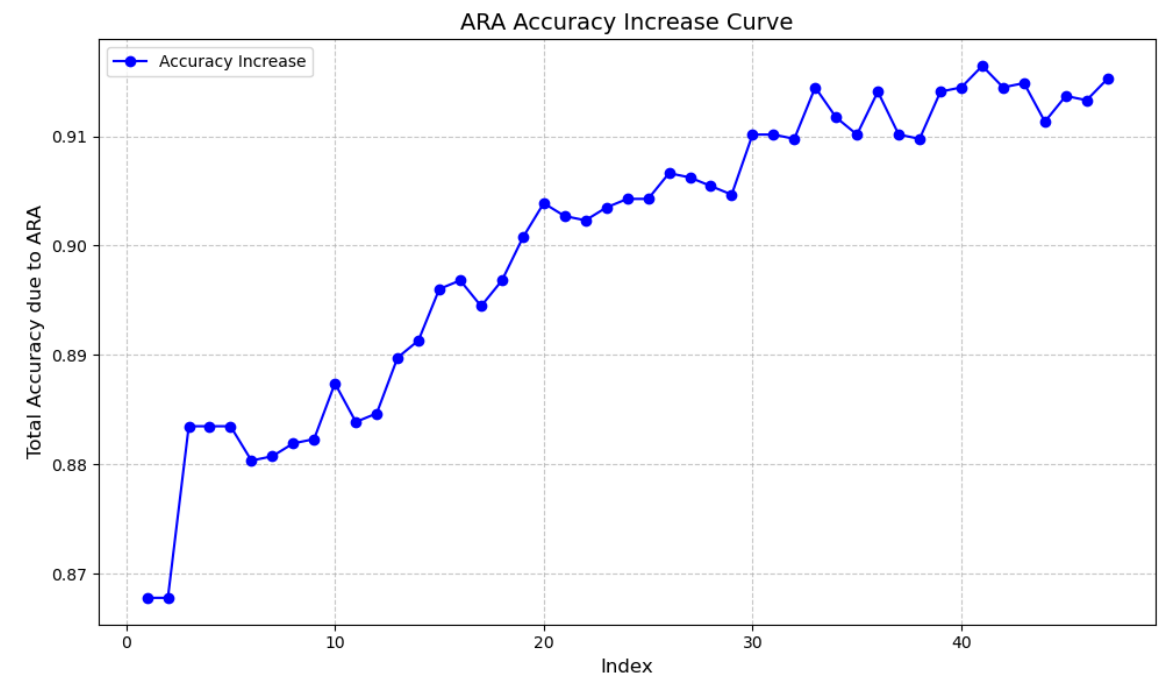


Graph 5 - Increase in model accuracy for the ARA method as attributes are incrementally added to the dataset (own processing)



For the purpose of comparing the two methods, it was again only necessary to work with the first 47 attributes for the ARA method.

Graph 6 - Increase in model accuracy for the ARA method as the first 47 attributes are incrementally added to the dataset (own processing)



It can be noticed that in both methods, the accuracy increased as the attributes were added. Again, however, the increase is not gradual, with decreases in accuracy at various points.

It is also important to note that for both methods, although the methods provided the 47 and 98 most important attributes of the dataset for SHAP and ARA (47 in the case of ARA for later comparability), there is only an increase of approximately 6-7% across the process. It is worth mentioning, that the entire dataset contained over 1600 attributes and thus the most important attributes for either method represented only a fraction of the size of the original dataset.

The numerical integration results for the area under the curve calculation was equal to approximately 42,005 for the SHAP method and approximately 41,414 for the ARA method for the 47 attributes. The difference in area under the curve is small for both methods. If we stick to the hypothesis that a smaller area under the curve means a better result, since an extremely good method would lead to a significant drop and a smaller area right at the beginning of the process, and conversely a very bad method would keep the same level of accuracy and thus a larger area under the curve, then the ARA method has a better result here.

### **3.4.3 Summary of method comparisons for incremental attribute addition**

The aim of the procedure described in this chapter was to re-examine how the results of the feature importance assessment methods differ, compare the two methods and assess whether either method can be used to interpret the model. Based on the measurements, it can be concluded that both methods have similar results. Compared to the previous procedure, the accuracy range for the first and last attribute examined is larger. However, it is still not sufficient to conclude that either method can be used for model interpretation.

However, at the same time, based on this and the previous procedure, we can assume that due to the jumps in accuracy that occurred during the measurement process, the attributes of the dataset are correlated. From the perspective of the research questions, correlation can be considered as a factor negatively affecting the results of the methods, i.e. it relates to the first secondary research question. The investigation of the influence of attribute correlations is the subject of further follow-up chapters.

Table 2 - Measurement results for the ARA/SHAP stepwise addition method (own processing)

| Method | Accuracy for the first attribute | Accuracy for the last attribute | Area under the curve |
|--------|----------------------------------|---------------------------------|----------------------|
| SHAP   | 0.86779129                       | 0.92899176                      | 42.0054              |
| ARA    | 0.86779129                       | 0.91526088                      | 41.4140              |

### 3.5 Summary of the comparison of SHAP and ARA methods using dataset adjustment

When comparing the SHAP and ARA methods by adjusting the datasets and subsequent AOPC analysis, it was found that neither procedure could define with absolute certainty a method that provided a sufficient explanation of the importance of each attribute, and thus it was not possible to explain on the basis of which attributes or how the trained model achieved its accuracy.

For both procedures, i.e., successively removing attributes according to their importance according to the SHAP or ARA method, training the model and measuring accuracy, and/or adding attributes to the dataset according to their importance and then training and measuring the accuracy of the model, results were obtained from which it can be inferred that the attributes of the dataset are most likely strongly correlated, making it impossible to unambiguously identify the attributes whose removal or addition would lead to a fundamental change in the accuracy of the model. Given this, for such a correlated dataset with such a robust predictive model, it is not possible to unambiguously explain the prediction of the model using the feature importance method SHAP or the ARA method.

In terms of the research questions, the procedures described above were primarily concerned with the main research question: how does the feature importance assessment of the SHAP and ARA methods differ for the CatBoots black-box model, and which method provides better results for model interpretation?

Based on the procedures described in the previous chapters, we can say that we obtain similar results for both methods in the measurements. Based on these measurements, it also turns out that neither method can be used for model interpretation. This is because both methods rank the attributes in order of importance such that removing or adding them does not result in a large/substantial change in the accuracy of the model.

At the same time, it was found through the procedures used that the attributes were likely correlated, which would explain the jumps in model accuracies when the dataset was

adjusted. Correlation can be seen as a negative factor affecting the results of the methods for feature importance. This observation relates to the first secondary question and the investigation of the effect of correlation is the subject of the following chapters and procedures.

## **3.6 Dataset modification using Factor and Principal Component Analysis**

Since the results of the previous testing suggested that the attributes in the dataset were highly correlated, which would preclude an effective explanation of the model predictions purely on the dataset attributes, the next step was to conduct a factor analysis and principal component analysis.

The aim of performing factor analysis and principal component analysis on the dataset was to obtain output in the form of components or factors, which would then be used to unambiguously determine whether the removal or addition of attributes belonging to a given factor or component would have a significant impact on the accuracy of the model and whether, if factors or components are gradually removed, there will be a gradual decrease in the accuracy of the model and thus the ability to interpret the behaviour of the prediction model based on these components/factors.

In this part of the work, the measurement was extended by another method. In addition to the SHAP and ARA methods, the permutation feature importance method was also used. Specifically, this was the `permutation_importance` method from the `scikit-learn` library.

The content of this part of the thesis mainly the first and second secondary research questions. Namely, whether there are factors present in the dataset that negatively affect the results of the methods. And also whether the results will change if the negative factors are removed.

However, it is understandable that the results of this part of the thesis can be related to the main research question itself. If negative factors are present and then removed how will the results of the methods differ and can any of them be used to interpret the model?

### **3.6.1 Dataset preparation for factor analysis**

Before conducting a factor analysis and confirming the hypothesis that the dataset attributes are correlated, it was necessary to conduct tests prior to factor analysis—specifically, the KMO and Bartlett's test. The tests were conducted on the original dataset after removing the target variable. This dataset was initiate as `data_df_clean_factor`.

Image 3 - Performance of KMO and Bartlett's test and their results (own processing)

```
#test for factor analysis - Bartlett's test and KMO test - values sufficient
bartlett_test = calculate_bartlett_sphericity(data_df_clean_factor)
kmo_all, kmo_model = calculate_kmo(data_df_clean_factor)
print(f"Bartlett's Test: {bartlett_test}")
print(f"KMO: {kmo_model}")

C:\Users\Marek H\anaconda3\Lib\site-packages\factor_analyzer\factor_analyzer.py
    statistic = -np.log(corr_det) * (n - 1 - (2 * p + 5) / 6)
C:\Users\Marek H\anaconda3\Lib\site-packages\factor_analyzer\utils.py:244: User
e to its determinant being at or very close to zero.
    warnings.warn(
Bartlett's Test: (inf, 0.0)
KMO: 0.8113517945912486

data_df_clean_factor.head(5)
```

The Bartlett test for the original dataset resulted in (inf, 0.0).

Infinity in Bartlett's test means that the determinants of the correlation matrix are very close to zero. In practice, this means that the dataset contains very strongly correlated values or directly linear dependencies. In general, in this case, the matrix is almost singular and this can lead to numerical problems in further analyses, for example in factor analysis. (Matematická biologie, n.d.)

The KMO test result for the unadjusted dataset was equal to approximately 0.811. Thus, both tests showed that the dataset contains strongly correlated values. However, Bartlett's test additionally found that the dataset may contain some very strongly correlated or even linear relationships. This is an important finding for further subsequent analyses. For the purpose of factor analysis, the dataset had to be modified so that the Bartlett test result was not infinite. This could be achieved by removing some strongly correlated values from the dataset. (Statistic How To, n.d.)

The dataset was adjusted for factor analysis in two ways. The first method involved removing highly correlated values. It should be noted that, for the purposes of later analysis, this procedure did not remove those attributes of the dataset that were identified as being the most important by feature importance according to the SHAP and ARA methods.

This first procedure proceeded as follows. An absolute correlation matrix was calculated, then an upper triangular matrix was created. Next, highly correlated columns were searched for by finding an index in the correlation matrix for each column, using the mask of the upper triangular matrix to compare pairs of values only once, and checking whether a column had a correlation greater than a set threshold. This was eventually set to 0.4. After testing, this value was the best value to remove enough values without affecting the dataset so much that factor analysis could no longer be performed. Strongly correlated columns were then removed. Predefined functions were used for the procedure.

For this setting, 613 correlated values were removed.



Image 4 - Removal of strongly correlated values (own processing)

```
# Dataset modification -correlation
# Correlation matrix
corr_matrix = data_df_clean_factor.corr().abs()

# Upper triangular part of the nut
upper_triangle = np.triu(np.ones(corr_matrix.shape), k=1)

# List of columns that have a strong correlation (correlation > 0.4) with other columns
drop_cols = []

# Iteration over all columns in the correlation matrix
for column in corr_matrix.columns:
    col_idx = corr_matrix.columns.get_loc(column)

    # Creating a logical mask for the values in the upper triangular part
    mask = upper_triangle[col_idx] == 1 # Getting the values in the upper triangular part for the given column

    # Check for correlations greater than 0.4
    if any(corr_matrix[column][mask] > 0.4) and column not in array_ara_importance and column not in array_shap_importance:
        drop_cols.append(column)

print(f"removing {len(drop_cols)} correlated attributes...")

# remove columns from the original dataset
data_df_clean_factor = data_df_clean_factor.drop(columns=drop_cols)

removing 613 correlated attributes...
```

The second procedure involved removing columns with too little variance. The assumption here was that columns with too little variance, in this case, columns with zero values, would be highly correlated with each other and may contribute to numerical problems in factor analysis.

This procedure involved obtaining the variance for each column, ordering the variances in ascending order, and then removing those values that fell within a certain percentile. This value was set to the 1st quartile. In other words, the quarter of attributes with the lowest variance was removed.

For this setting, 268 attributes were removed.

After these two processes for adjusting the dataset, the Bartlett and KMO tests were again performed to verify whether there was an improvement in the Bartlett test and whether the dataset was still suitable for factor analysis after adjustment.

The test results were approximately 0.7987 for the KMO test and (4217484.591005148, 0.0) for the Bartlett test. Thus, both tests showed that it was still possible to perform factor analysis, and at the same time, the infinity problem for the Bartlett test was solved using the above procedures.

Image 5 - Performance of KMO and Bartlett's test and their results after procedures which reduced correlation of attributes (own processing)

```
#test for factor analysis - Bartlett's test and KMO test - values sufficient
bartlett_test = calculate_bartlett_sphericity(data_df_clean_factor)
kmo_all, kmo_model = calculate_kmo(data_df_clean_factor)
print(f"Bartlett's Test: {bartlett_test}")
print(f"KMO: {kmo_model}")

C:\Users\Marek H\anaconda3\Lib\site-packages\factor_analyzer\utils.py:244: UserWarning:
e to its determinant being at or very close to zero.
  warnings.warn(
Bartlett's Test: (4217484.591005148, 0.0)
KMO: 0.7987393616243856
```

### 3.6.2 Factor analysis

After adjusting the dataset and performing Bartlett and KMO tests, the next step was to perform the factor analysis itself.

The procedure chosen for the factor analysis was as follows. First, the factor analysis was conducted once and the number of factors was set with a parameter of 20. The results of the factor analysis performed were to be visualised and the elbow that would correspond to the ideal number of factors was to be identified using a graph. Then the factor analysis was to be performed again but with the ideal number of factors. However, after conducting the initial factor analysis and visualizing its result, it turned out that the number of factors was not equal to the number set in the parameter, i.e. 20, but was equal to the number of attributes in the dataset.

To solve this problem, a procedure was chosen in which PCA was performed before factor analysis. The aim was to reduce dimensionality and suppress multicollinearity. The procedure for the factor analysis remained the same, i.e., conduct the PCA, then conduct the initial factor analysis, identify an optimal number of factors, and conduct the factor analysis again.

However, the implementation of this procedure showed that performing PCA before factor analysis and subsequent factor analysis does not solve the problem. When performing the PCA analysis, the number of components parameter was set to 100. The subsequent factor analysis worked with the output of the previous PCA, but despite the fact that the number of factors parameter was set to 20, the factor analysis worked with the number of factors corresponding to the number of components from the PCA.

It should also be mentioned that due to the fact that we were working with a highly correlated dataset, it could be assumed that the ideal number of components or factors would be rather lower and the original parameters set for the number of components or factors in the initial PCA or factor analysis were high enough.

Image 6 - Performing PCA before factor analysis (own processing)

```
#PCA
pca = PCA(n_components=100)
pca.fit(data_df_clean_factor)
ev = pca.explained_variance_
pca_components = pca.transform(data_df_clean_factor)
```

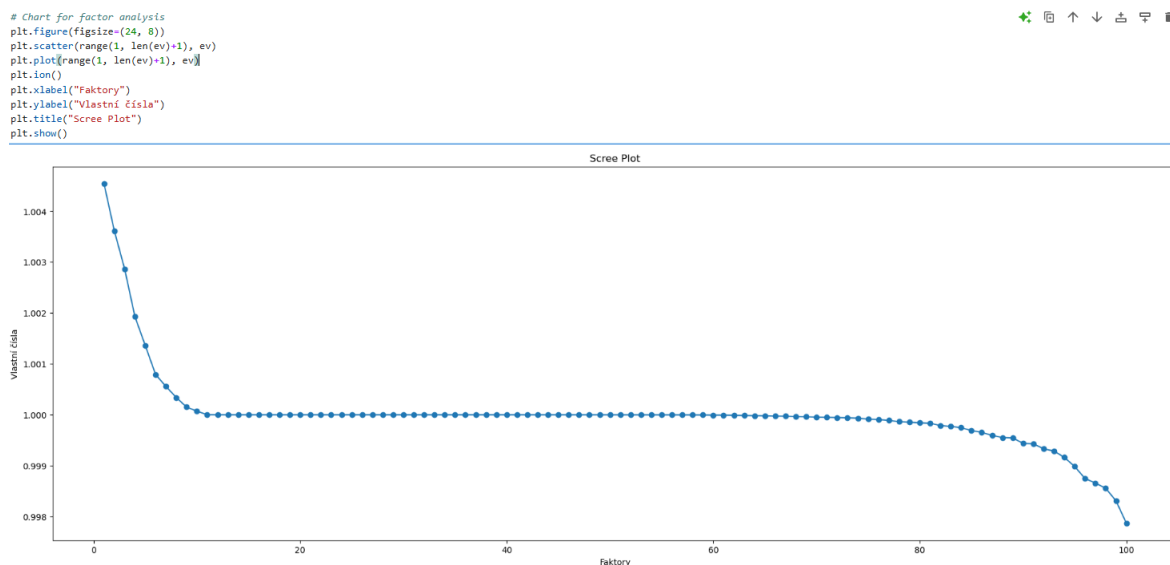
Image 7 - Perform factor analysis on the PCA output with the number of factors set to 20 (own processing)

```
#factor analysis
# max number of factors
factors = 20

faktor_an = FactorAnalyzer(n_factors=factors, rotation="varimax")
faktor_an.fit(pca_components)

#factor eigenvalues
ev, v = faktor_an.get_eigenvalues()
print(f"Vlastní čísla pro faktory: {ev}")
```

Graph 7 - A graph of the factor analysis output showing that the number of factors is equal to the number of PCA components (own processing)



### 3.6.3 Principal Component Analysis

The previous procedure showed that if a PCA is performed on a dataset and a factor analysis is performed on its output, the result will be that the factor analysis will work with the same number of factors as the number of components set for the PCA. Given this fact, the procedure has been modified. Instead of the factor analysis, only the PCA was performed, with the output of the PCA being followed by other procedures and measurements. Thus, the next subsections of this thesis deal with the performance of PCA and the description of these mentioned procedures on the output of the PCA analysis.

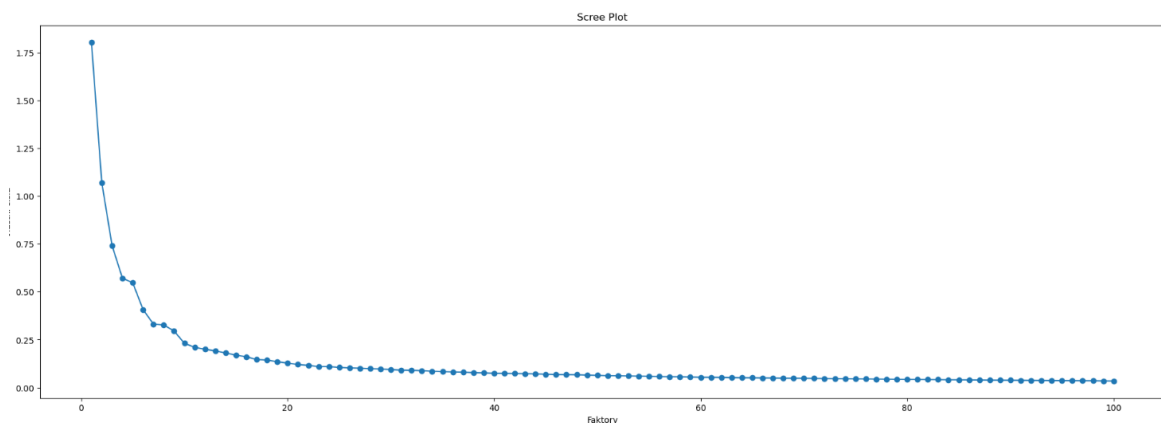
Since PCA could be performed without modifying the dataset by removing highly correlated or low variance attributes, PCA was performed on the original dataset to reduce the risk of bias in the results.

### 3.6.4 Performing principal component analysis

The first step for further possible investigation was to perform a PCA. This analysis was conducted in the way that factor analysis was originally intended to be conducted, i.e., an initial run of the analysis, identifying the optimal number of components, and a second run of the analysis with the parameter set to the ideal number of components.

After the initial analysis, the number of components was again set to 100. The results of this analysis were visualised.

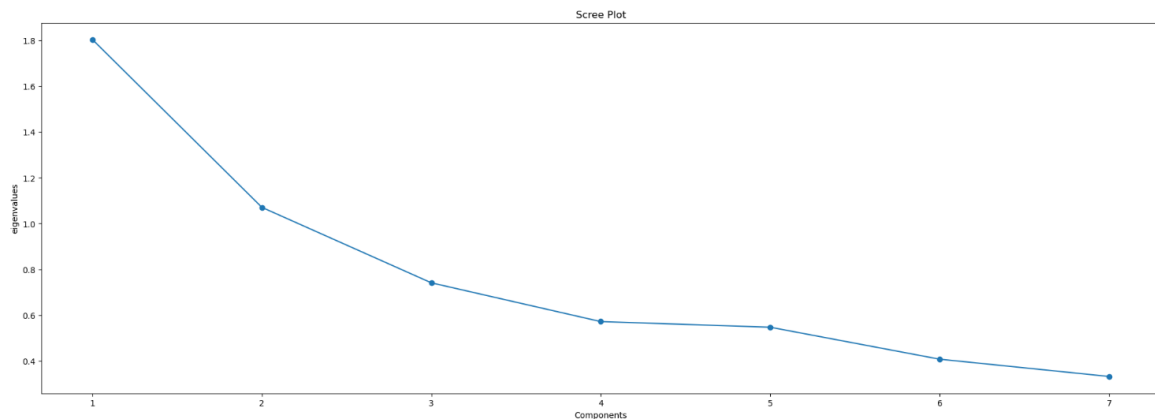
Graph 8 - Graph of the initial PCA analysis to identify the optimal number of components (own processing)



From the visualization of the results, it was possible to determine the elbow that corresponds to the ideal number of components. The optimal number of components according to the elbow was 7.

However, here it is important to look at the visualization of the PCA output for the optimal number of components. Looking at the graph, one can notice that only 2 components have eigenvalues greater than 1. So here one could argue that the ideal number of components is not 7 according to the elbow from the visualization of the initial PCA and say that the optimal number of components is 2 according to Kaiser's rule. However, the problem here is that if we work with only 2 components, then due to the nature of the other procedures, it will not be possible to explain the behaviour of the model since the attributes will be divided into only two components. So for further measurements, it is better if we work with 7 components as their ideal number.

Graph 9 - Graph showing the PCA output with the ideal number of components. The eigenvalues of the components are captured (own processing)



### 3.6.5 Attribute removal based on component membership and feature importance

After performing PCA on the dataset, it was possible to perform the first of the procedures on its output. The principle of this procedure was to take the ranked attributes of the original dataset according to their importance and determine to which component they belong. Once this was identified all attributes falling into that component were removed from the dataset. In contrast to the original measurement where attributes alone were removed, the aim here was to remove other correlated attributes and examine what impact this would have on the output of the prediction model.

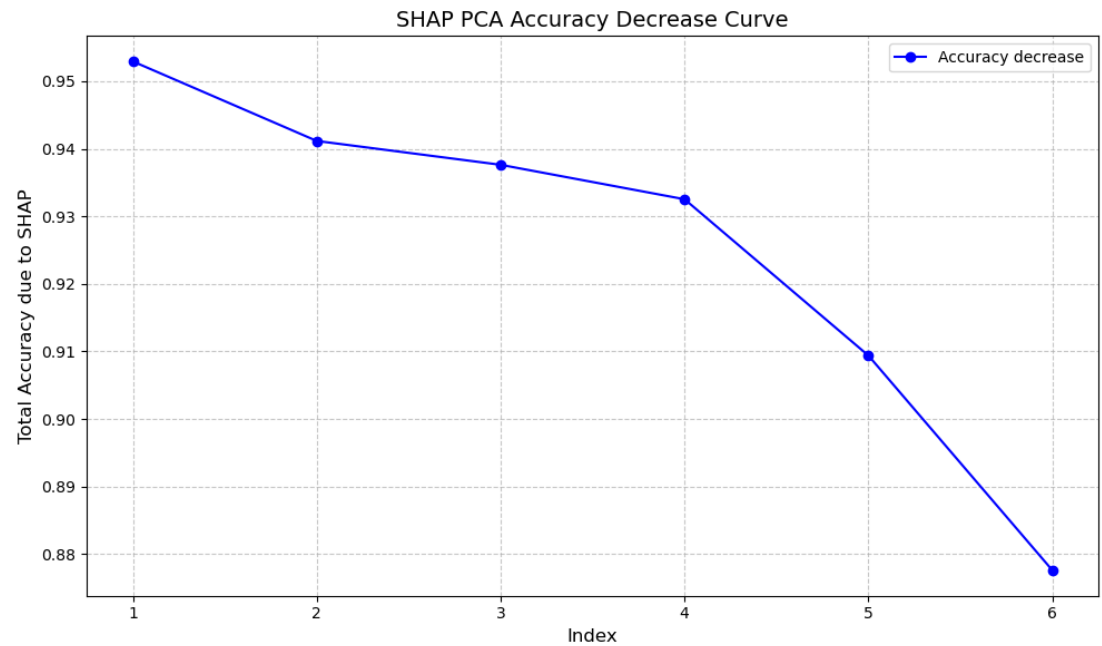
The specific procedure was as follows. A map was created, to which all components were added using a for loop, and for each component the number of attributes falling under that component was determined by loading each attribute.

Next, a while loop, an empty array to store the model accuracies, and a variable to browse the array of attributes according to their importance to the SHAP or ARA method were created.

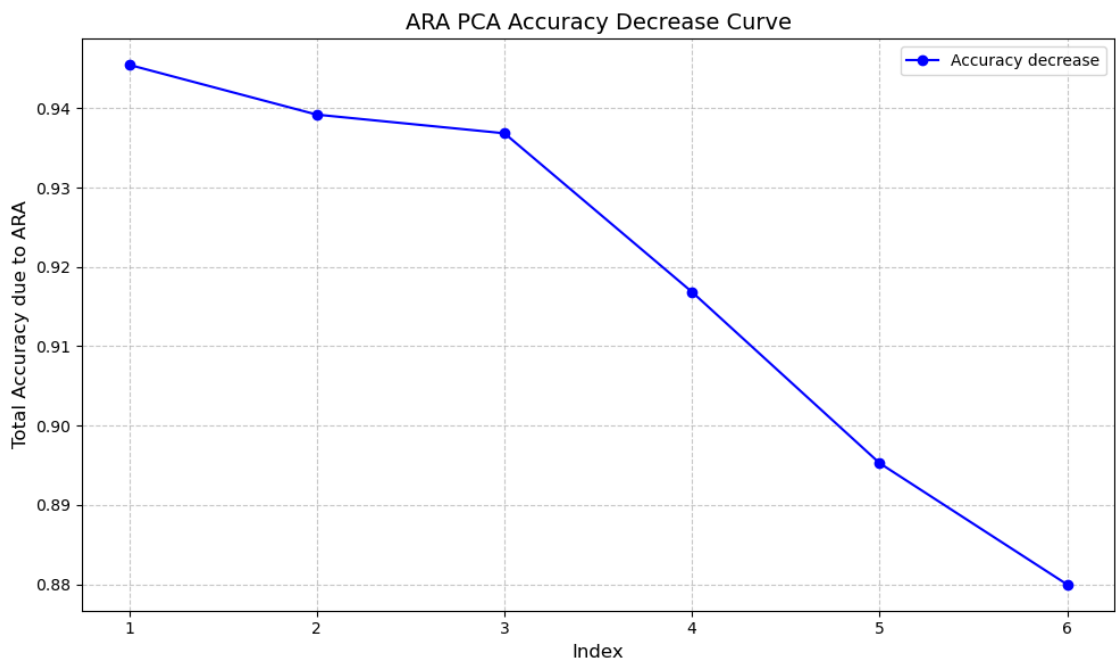
In each iteration of the while loop, one attribute was selected at a time, starting with the most important one. The loadings for this one attribute were determined and the component to which it belongs was also determined. If the component was still in the map and thus not removed, all attributes that belonged to that component were removed from the dataframe and the component was removed from the map. If the component had already been removed, the loop moved on to the next variable in the attribute list for the SHAP or ARA method, depending on its importance.

In each iteration of the cycle, if there was a change in the dataset, the model was re-trained on the modified dataset and its accuracy was measured. The accuracies of the models were visualised. The areas under the curve were calculated for both the SHAP and ARA methods.

Graph 10 - Accuracy of models in attribute removal as a function of their membership in PCA components for SHAP (own processing)



Graph 11 - Accuracy of models in attribute removal as a function of their membership in PCA components for ARA (own processing)



The areas under the curve are approximately 4.6359 for the SHAP method and approximately 4.6008 for the ARA method.

Again, if we work under the assumption that the method that better explains the importance of attributes has a steeper declining accuracy curve and hence a smaller area under the curve, then the ARA method performs better on this comparison.

Looking at both curves, it is important to note two differences. What both curves or methods have in common is that their decline is not linear but that at one point there is a more pronounced decline. In the SHAP method, this decrease occurs when 4 components are removed and in the ARA method when 3 components are removed.

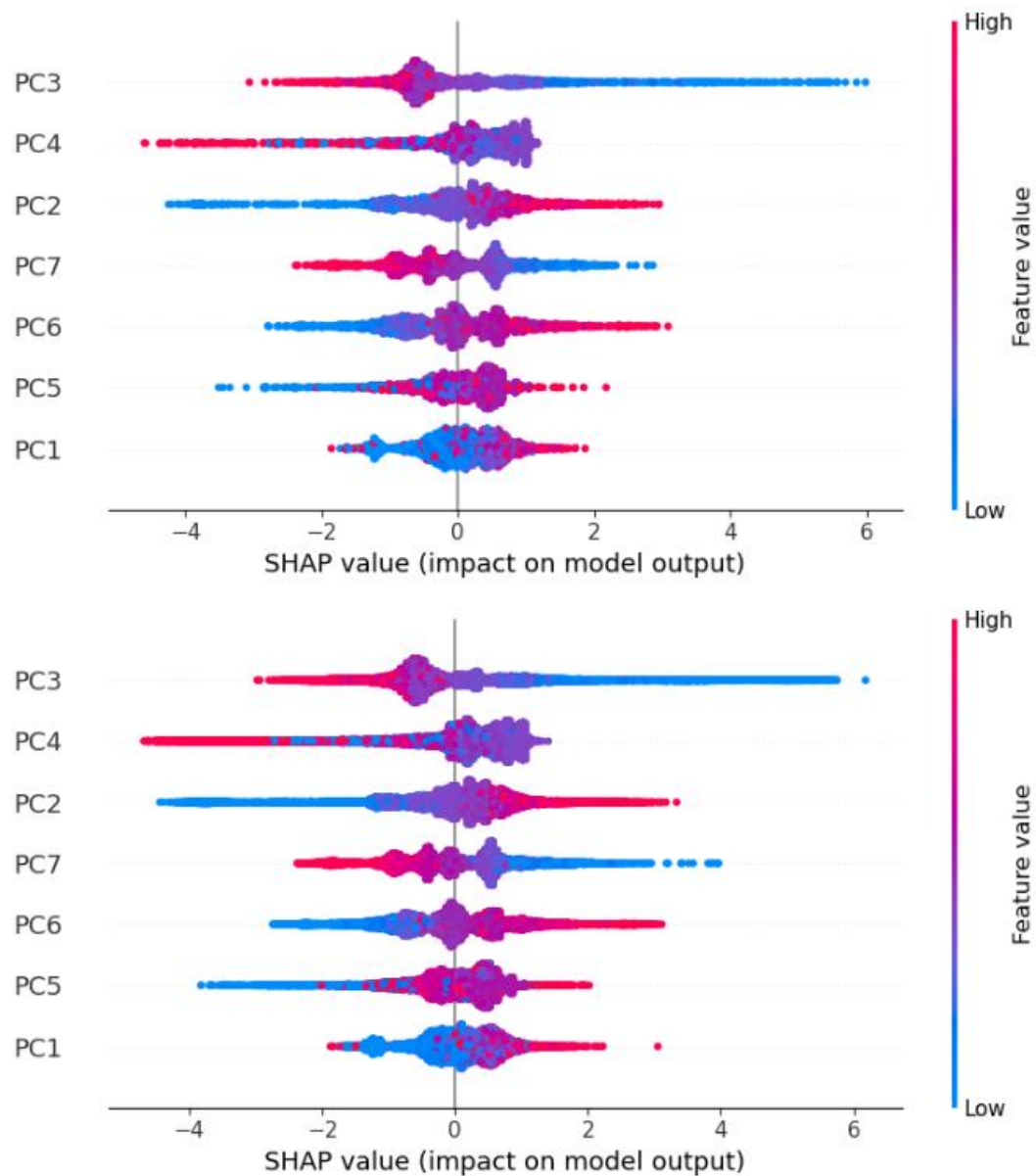
The procedure described above, i.e. using PCA and then removing attributes according to their importance, solves the problem that arose when removing the attributes themselves. There is now a gradual decrease in accuracy without an increase in accuracy at any point.

### **3.6.6 Performing SHAP and permutation feature importance method on the output of the PCA analysis**

The second procedure involved applying the above-mentioned methods, i.e. SHAP, ARA and permutation feature importance directly to the PCA output of the analysis. Thus, the aim was to obtain the importance of each component and then examine the impact of removing them. In this case, however, attributes would not be removed according to their importance but directly by component.

The application of the methods to the PCA output was as follows. After performing the PCA analysis, splitting the output into training, test and validation sets, the model was trained. After model training, predictions and accuracy evaluation, SHAP and permutation feature importance methods were performed respectively. For SHAP, this involved calculating SHAP values, plotting a SHAP summary plot for the test and training data. Then, the importance of each component was calculated. For the ARA method, since it is not a post hoc method, it was not necessary to run the model before using it.

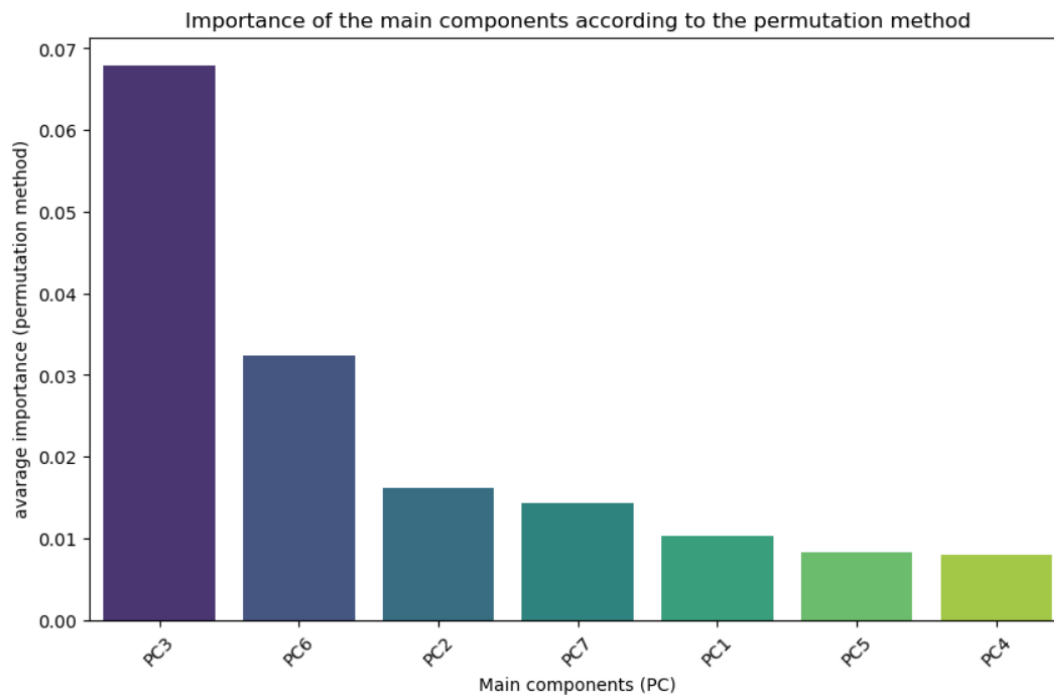
Graph 11 - Summary plot for the impact of each PCA component on the model output for the SHAP method (own processing)



For the permutation method, the procedure involved permutation importance analysis alone then storing the results and ranking them by importance followed by visualisation.



Graph 12 - Visualisation of the importance of individual PCA components according to the permutation method (own processing)



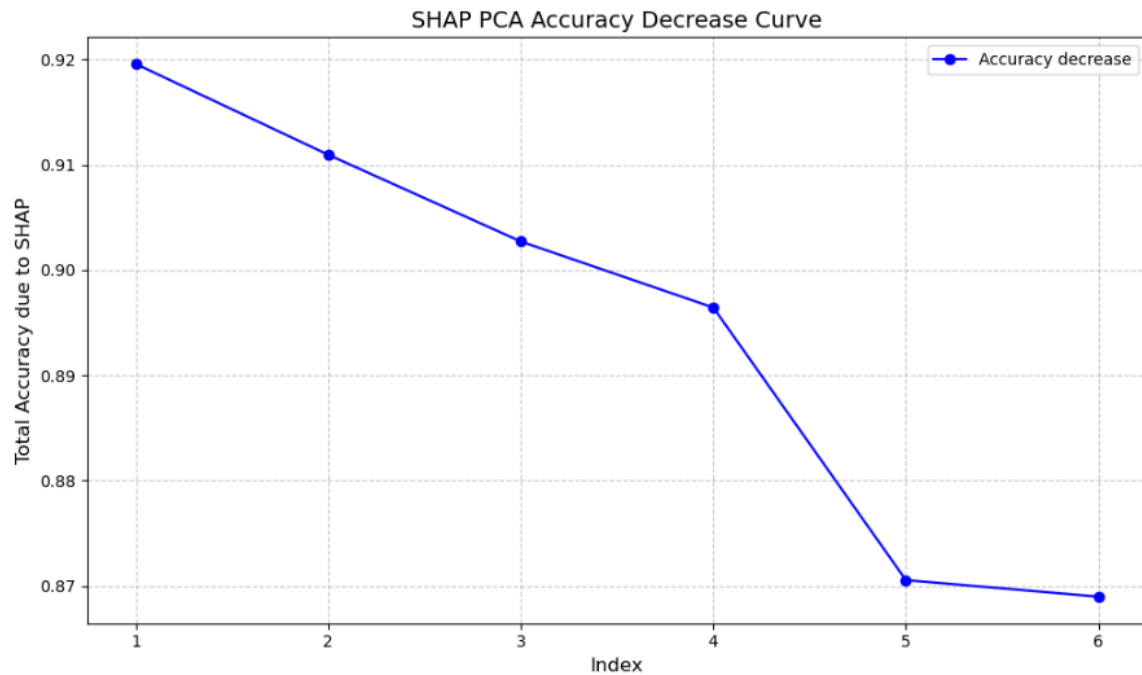
Although the order of importance of the components is different then in the case of SHAP method, both methods defined the most important component as PC3. The other rankings differ further for the two methods.

The next procedure after using both methods was to store the ranked components in order of importance. A for loop was then created to store the component names in an array.

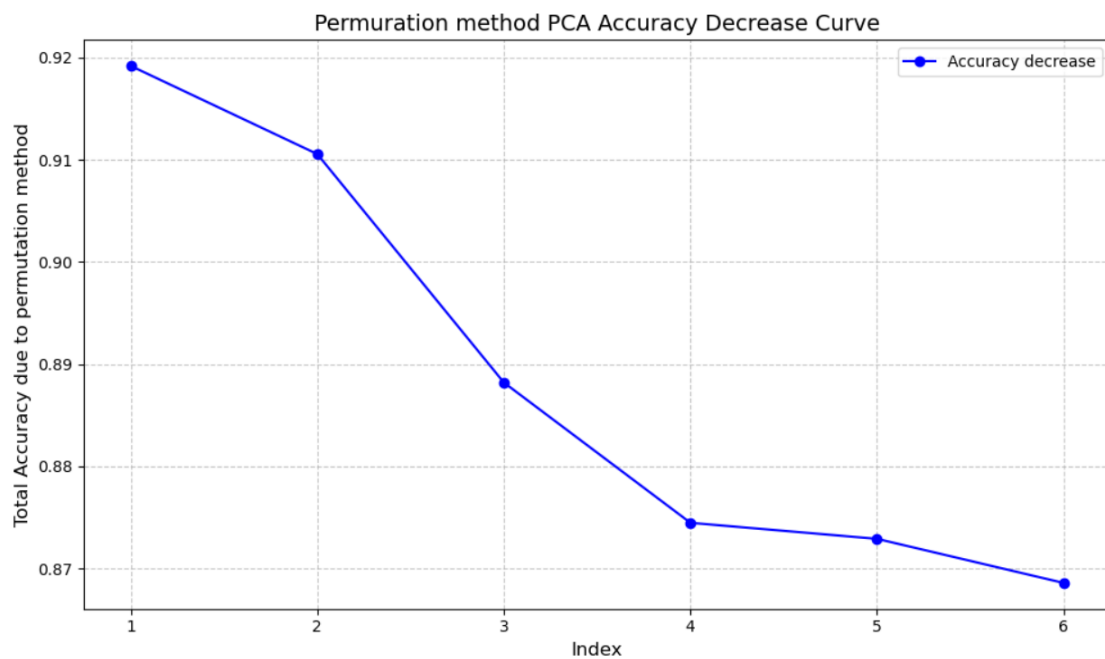
Next, an array was created for the accuracy of each model, and another for loop was created that traversed the array with the component names. At each iteration of the loop, the component was removed from the data set and the model was re-trained. Its accuracy was then recorded.

The output of this process was the accuracies of models trained on a dataset with sequentially sampled components. The individual accuracies were visualized.

Graph 13 - Model accuracies for sequential component removal using the SHAP method (own processing)



Graph 14 - Model accuracies for sequential component removal using the permutation method (own processing)



For both methods, the areas under the curve were calculated using numerical integration. For the permutation method, the area was equal to approximately 4.4399 and for the SHAP method approximately 4.4792.

If we stick to the assumption that the method that explains the importance of each component or attribute better has a smaller area under the curve, then the permutation method provides a better result in this case.

However, it is also interesting to look at the course of the curves themselves. For the SHAP method, there is again a more pronounced decrease after the removal of the 4 components. In this case, the components are PC3, PC2, PC4, PC7. However, in the case where the components were removed in order of attribute importance, the order of removal was PC1, PC6, PC5, PC2.

In both cases, when 4 components were removed, there was a decrease, despite the fact that neither the order nor the individual components corresponded.

In this case, the permutation method already shows a more gradual decrease in the removal of individual components.

For the mentioned procedure, it should be noted that although 7 components were worked with as the optimal number, the number of iterations within the cycle when removing them is equal to 6. This is due to the fact that it is not possible to remove all components, as then all attributes of the dataset would be removed and it would not be possible to re-train the model and measure its accuracy.

### **3.6.7 ARA method**

Due to the different nature of the ARA method and its results, this subsection is devoted to this method for the procedure involving PCA.

In contrast to the permutation and SHAP methods, the output of the ARA method is not the most important attributes or components, but specific intervals of values of a given attribute or component.

This fact makes it difficult to later assess the quality of the method output. In the previous two cases, the procedure used was to remove components from the dataset according to their importance using the permutation and SHAP methods, respectively. In this case, however, it is not possible to remove the entire component as the ARA method works with sub-intervals within a single component.

Thus, for the purpose of output validation here, individual components were not removed, but the values that fell within a given interval in a given component were reset, the model was re-trained and its accuracy was measured.

The procedure for this method was as follows. The first step was to perform a PCA analysis. This output was converted into a dataframe and the target variable was added back into the dataframe. This was removed from the original dataset as it could not be handled in the PCA framework. The output of this was therefore a dataset that contained 7 components as independent variables and the target variable.

However, for the purposes of the method, it was also necessary to convert the component values from continuous to categorical variables. In other words, each value of a particular record was converted to the interval into which it falls. The number of categories, was set to 3. Then the actual method was performed using the araxai library.

Image 8 - Code and results for the ARA method (own processing)

```
] : # Converting components to categorical variables
components = ['PC1', 'PC2', 'PC3', 'PC4', 'PC5', 'PC6', 'PC7']
for varname in ['PC1', 'PC2', 'PC3', 'PC4', 'PC5', 'PC6', 'PC7']:
    pca_df[varname] = pd.qcut(pca_df[varname], q=3,) # 3 categorie

# specification of target value and components
a = ara(df=pca_df,
        target='medium', # target value |
        target_class='medium:65',
        options={"max_depth":1})
        #CL= components)

# printing of results
a.print_result()
a.print_task_info()
a.print_statistics()

print(f"    ...full results : {a.get_task_info()}")
print(f"    ...only results : {a.get_results()}")
print(f"    ...only rules    : {a.get_rules()}")

-.. PC2((-3.0509999999999997, -0.31]) /2.7 (= /2.7)
+.. PC3((-3.995, -0.15]) x2.4 (=x2.4)
-.. PC3((-0.15, 0.0668]) /2.9 (= /2.9)
-.. PC3((-0.15, 0.0668] (0.0668, 4.086]) /3.1 (= /3.1)
-.. PC3((0.0668, 4.086]) /3.3 (= /3.3)
-.. PC6((-2.597, -0.153]) /2.9 (= /2.9)
-.. PC6((-2.597, -0.153] (-0.153, 0.113]) /2.3 (= /2.3)
+.. PC6((0.113, 3.677]) x2.1 (=x2.1)
```

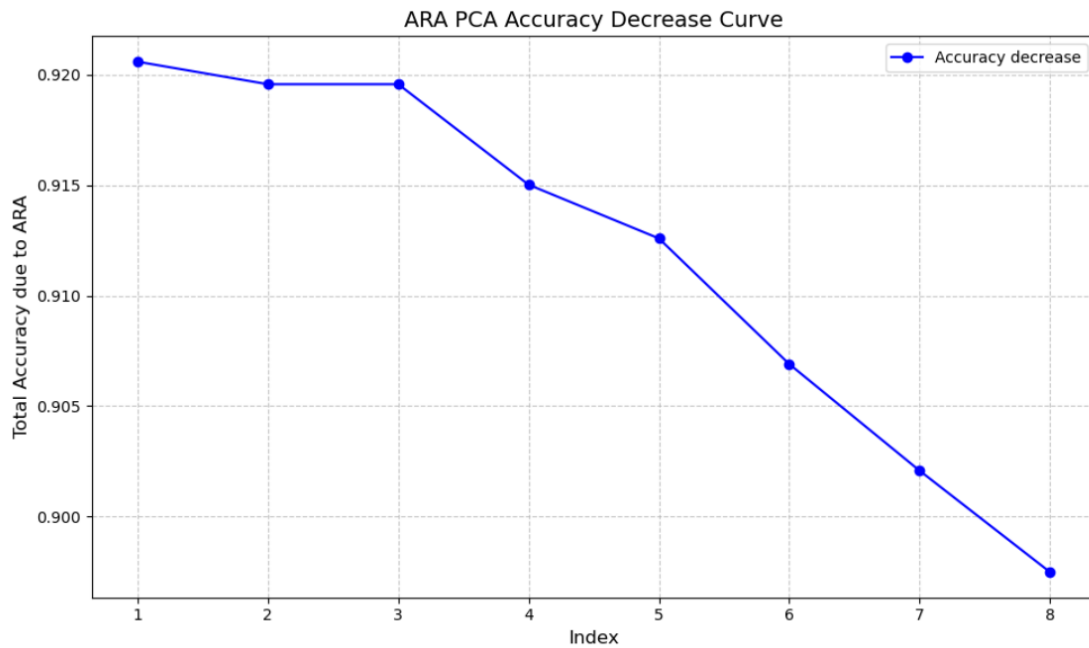
According to the ARA method, the most important intervals are within component 3. These intervals had the highest scores as shown in Image 8. Thus, a parallel can be seen here with the previous methods, i.e. SHAP and the permutation method, which also identified component PC3 as the most important component. However, the previous methods did not specify a specific most important range, i.e., the intervals within a given component.

Next, the procedure was followed in which the values of the components that fell into the given interval were gradually replaced by random values outside the interval in the dataset according to their importance according to the ARA method.

Initially, the values were reset instead of random substitution, but the output of this variant of the procedure was not sufficiently conclusive. Therefore, the random substitution option was finally used.

Since this procedure involves random substitution of values outside the interval. The above procedure was repeated 10 times and the measured values were averaged.

Graph 15 - Average values of accuracies for random interchange of values outside the interval according to the ARA method (own processing)



The output of the above procedure can be seen in Graph 15. It can be seen that there is a gradual decline. The overall magnitude of the decrease is about 3%. However, it should be mentioned that in this method, the whole components are not removed but only parts/intervals were randomly swapped, unlike the permutation method and SHAP method where the whole components were removed during the sampling process until only one component was left in the dataset.

### 3.7 Summary of the results of PCA and factor analysis

Using PCA and factor analysis, or Bartlett's and KMO tests, the assumption that arose when examining the impact of removing attributes from the dataset was confirmed, namely that the attributes of the dataset are highly correlated. The initial intention to apply factor analysis to the dataset was not effective even when highly correlated attributes and attributes with high variance were removed from the dataset. When PCA was used before factor analysis, it was found that factor analysis works with a number of factors corresponding to the set number of PCA components, so the procedure was modified and only PCA was used.

The ideal number of components was identified from the visualization as 7, despite the fact that only 2 components had eigenvalues greater than 1. The reason for not using only 2 components was that using only 2 components would make it impossible to effectively measure or explain the prediction model, as the attributes would only be divided into 2 groups.

The 2 procedures were applied to the PCA output. The results of the first procedure of removing components according to the original distribution of attributes according to importance and the second procedure of removing components according to their importance according to SHAP, ARA and permutation method showed that the decrease in accuracy after PCA was more obvious, with no increase in accuracy during measurement, and are therefore better for the ability to interpret the model than removing attributes from the dataset alone.

For all three methods, it can be noticed that there is a greater decrease in accuracy at a certain point in the first procedure and in the SHAP method as well as in the second procedure.

As mentioned in the introduction, this part of the thesis deals mainly with secondary research questions, but in effect the main research question can be answered based on the output of this part.

The first secondary question concerned the presence of factors negatively affecting the results of the methods. At the beginning of this part of the thesis, KMO and Bartlett's test were conducted and confirmed the presence of correlation between attributes. The measurements that followed helped to answer whether correlations do indeed have a negative impact on the results of the methods. The measurements showed that they did. The procedures used that were performed on the PCA output led not only to a greater decrease in accuracy but more importantly to their gradual decreases in accuracy. Based on these facts, it can be concluded that factors with negative impacts on the results of the methods were present. At the same time, it was measured that when these factors, i.e. correlation, were removed, there were clearer and more pronounced decreases in accuracy and therefore better interpretability of the model based on the methods used.

Thus, this answered the secondary research questions. In the context of the main research question, it can be concluded for this chapter or part of the thesis that neither method provides a significantly better assessment of the feature importance of the individual components. The conclusions are therefore similar. The range of accuracies here, while greater than in the case of dataset modification using simple attributes, is not sufficient to interpret the model based on these approaches.

Table 3 - The importance of each component according to the 3 methods used (own processing)

|                    | <b>1. = The most important</b>  | <b>2.</b>                                       | <b>3.</b>                       | <b>4.</b>                                 | <b>5.</b>                       | <b>6.</b>                        | <b>7. = The least important</b> |
|--------------------|---------------------------------|-------------------------------------------------|---------------------------------|-------------------------------------------|---------------------------------|----------------------------------|---------------------------------|
| SHAP               | PC3                             | PC4                                             | PC2                             | PC7                                       | PC6                             | PC5                              | PC1                             |
| Permutation method | PC3                             | PC6                                             | PC2                             | PC7                                       | PC1                             | PC5                              | PC4                             |
| ARA                | PC3 in interval (0.0668, 4.086] | PC3 in interval (-0.15, 0.0668] (0.0668, 4.086] | PC3 in interval (-0.15, 0.0668] | PC2 in interval (-3.05099, 999997, -0.31] | PC3 in interval (-3.995, -0.15] | PC6 in interval (-2.597, -0.153] | PC6 in interval (0.113, 3.677]  |

Table 4 - Model accuracies for selected methods (own processing)

| <b>Method</b>                                                | <b>Accuracy removing the first component</b> | <b>Accuracy removing the last component</b> | <b>Area under the curve</b> |
|--------------------------------------------------------------|----------------------------------------------|---------------------------------------------|-----------------------------|
| SHAP - component removal by feature importance of attributes | 0.95292271                                   | 0.87759906                                  | 4.6359                      |
| ARA - component removal by feature importance of attributes  | 0.94546881                                   | 0.87995292                                  | 4.6008                      |
| SHAP                                                         | 0.9195763                                    | 0.86896822                                  | 4.4792                      |
| ARA                                                          | 0.92310710                                   | 0.89682228                                  | 6.3713                      |
| Permutation method                                           | 0.9192                                       | 0.8686                                      | 4.4399                      |

# Conclusion

From the accuracy measurements for both methods it means SHAP and ARA, without attribute returns, it can be observed that there is a decrease in accuracy in both cases. However, in both cases, the decrease is not gradual, with increases in accuracy at some points. Also, in both cases, the accuracies of the models before and after the removal of all attributes ranked by importance according to both methods show only a small decrease, namely in tenths of percent or at most units of percent.

Based on the measurements described in the chapter of measurements for both methods without attribute returns, while it is possible to compare the two methods, given the small decrease in accuracy between the first and last attribute for both methods, it can be concluded that in the case of both methods, if used as a tool for model interpretation, their results will be ambiguous.

The procedure of chapter with incremental adding attributes supported the suspicion that the dataset attributes are correlated. This can be seen in the progression of the accuracy curve on which larger jumps in accuracy occur at certain points. It can also be noticed that the difference in accuracies at the beginning and end of the process is more significant than the first procedure.

It could therefore be assumed that if correlation could be removed as a factor negatively affecting the outcome of the methods, then better results could be measured and possibly either or both methods could be used to interpret the prediction model. In other words, it could be assumed that if we use e.g. PCA or factor analysis and test the methods on the output of these analyses, then the outputs will be better than for attributes alone. However, it should also be mentioned that using these analyses would impair the interpretability of the methods to explain the importance of attributes as a tool to explain the black box model.

Overall when comparing the SHAP and ARA methods by adjusting the datasets and subsequent AOPC analysis, it was found that neither method produces significantly better results than the other. Both methods had very similar results in the measurements.

For both procedures, i.e., successively removing attributes according to their importance according to the SHAP or ARA method, training the model and measuring accuracy, and/or adding attributes to the dataset according to their importance and then training and measuring the accuracy of the model, results were obtained from which it can be inferred that the attributes of the dataset are strongly correlated, making it impossible to unambiguously identify the attributes whose removal or addition would lead to a fundamental change in the accuracy of the model. Given this, for such a correlated dataset with such a robust predictive model, it is not possible to unambiguously explain the prediction of the model using the feature importance method SHAP or the ARA method.



Using PCA and factor analysis, or Bartlett's and KMO tests, the assumption that arose when examining the impact of removing attributes from the dataset was confirmed, namely that the attributes of the dataset are highly correlated. The initial intention to apply factor analysis to the dataset was not effective even when highly correlated attributes and attributes with high variance were removed from the dataset. When PCA was used before factor analysis, it was found that factor analysis works with a number of factors corresponding to the set number of PCA components, so the procedure was modified and only PCA was used.

The ideal number of components was identified from the visualization as 7, despite the fact that only 2 components had eigenvalues greater than 1. The reason for not using only 2 components was that using only 2 components would make it impossible to effectively measure model performance by one method or the other, as the attributes would only be divided into 2 groups.

The 2 procedures were applied to the PCA output. The results of the first procedure of removing components according to the original distribution of attributes according to importance and the second procedure of removing components according to their importance according to SHAP, ARA or permutation method showed that the decrease in accuracy after PCA was more obvious, with no increase in accuracy during measurement, and are therefore better for the ability to interpret the model than removing attributes from the dataset alone.

For both methods, it can be noticed that there is a greater decrease in accuracy at a certain point in the first procedure and in the SHAP method as well as in the second procedure. In the case of the ARA method it was a decrease after the removal of the 6th interval.

The procedures, analyses and measurements used in this thesis show that none of the methods, i.e. SHAP, ARA or the permutation method, perform significantly better than the other methods. This is the case even when working with independent i.e. uncorrelated attributes.

Although it could have been assumed during the measurement or data analysis that it would be possible to use the methods for black-box model interpretation, their use on PCA showed that as an interpretative tool, these methods are limited. In both cases, there was not a sufficient decrease in accuracy despite the removal of most attributes. Some of the most important groups of attributes can be described based on these methods, but the overall interpretation of the model is difficult with them. This applies to all the methods examined.

When the dataset was modified using the original attributes, the difference in model accuracy was in the tens or at most low units of percent. In the case of using PCA, i.e. removing correlations, the range of accuracy decrease was up to 10%.

This work can be followed up with an analysis of why the accuracy of the model is high despite the removal of most attributes from the dataset and whether this could be used to interpret the model.

Based on the results of this thesis, it is possible to answer the stated research questions:

- How does the feature importance assesment of the SHAP and ARA methods differ for the CatBoost black-box model, and which method provides better results for model interpretation?

Based on the procedures in which the accuracy of the models was measured when modifying the dataset using either attributes or components as PCA output, it can be concluded that both methods had very similar results. The differences in measurements for both methods were negligible. The results of both methods show that their use as an interpretive tool for a given prediction model can be difficult and ambiguous.

- First secondary research question: Are here any factors that might negatively influence these results?

The results of the procedures where the dataset was modified using attributes suggested that there may be a negative factor affecting the results of the methods in the form of correlation between attributes. It could be assumed that the potential correlation prevents the SHAP or ARA method from effectively explaining the model, as the adjustment of the dataset did not sufficiently change the model output or decrease the accuracy due to the correlation.

The assumption of the influence of correlation was confirmed in the next part of the paper, specifically using Bartlet and KMO tests. These tests showed beyond doubt that the dataset attributes are strongly correlated.

The assessment of the negative effect of correlation on the evaluation of the importance of attributes was the main focus of the second part of this thesis. It showed that in the presence of correlation, feature importance methods are not able to assess with sufficient accuracy which attributes are relevant to the model in terms of their importance.

- Second secondary research question: How will the results of the methods change if factors negatively affecting results removed?

This question is answered by the procedure in which PCA was used. The aim of using PCA was to remove the demonstrated correlation between attributes and measure the change in model accuracy. Two procedures were performed to output the PCA. Removing components based on the membership of the original most important attributes to the components. The second procedure was to use SHAP, ARA and permutation method to determine the importance of each component. Then the components were removed and the decrease in accuracy was measured.

For both procedures, a gradual decline was measured and observed without large fluctuations. Based on these measurements, it can be concluded that removing correlation

as a negative factor improves the results of the methods. However, again, the decrease in accuracy is not sufficient to allow any method to be used as a reliable model interpretation tool even when the negative effect of correlation is removed.

# List of graphs

|                                                                                                                                  |    |
|----------------------------------------------------------------------------------------------------------------------------------|----|
| GRAPH 1 - DECREASE IN MODEL ACCURACY FOR THE SHAP METHOD WITHOUT RETURNING ATTRIBUTES..                                          | 22 |
| GRAPH 2 - DECREASE IN MODEL ACCURACY FOR THE ARA METHOD WITHOUT RETURNING ATTRIBUTES ...                                         | 23 |
| GRAPH 3 - DECREASE IN MODEL ACCURACY FOR THE ARA METHOD WITHOUT RETURNING ATTRIBUTES FOR THE FIRST 47 ATTRIBUTES.....            | 25 |
| GRAPH 4 - INCREASE IN MODEL ACCURACY FOR THE SHAP METHOD AS ATTRIBUTES ARE INCREMENTALLY ADDED TO THE DATASET.....               | 27 |
| GRAPH 5 - INCREASE IN MODEL ACCURACY FOR THE ARA METHOD AS ATTRIBUTES ARE INCREMENTALLY ADDED TO THE DATASET.....                | 28 |
| GRAPH 6 - INCREASE IN MODEL ACCURACY FOR THE ARA METHOD AS THE FIRST 47 ATTRIBUTES ARE INCREMENTALLY ADDED TO THE DATASET.....   | 28 |
| GRAPH 7 - A GRAPH OF THE FACTOR ANALYSIS OUTPUT SHOWING THAT THE NUMBER OF FACTORS IS EQUAL TO THE NUMBER OF PCA COMPONENTS..... | 35 |
| GRAPH 8 - GRAPH OF THE INITIAL PCA ANALYSIS TO IDENTIFY THE OPTIMAL NUMBER OF COMPONENTS...                                      | 36 |
| GRAPH 9 - GRAPH SHOWING THE PCA OUTPUT WITH THE IDEAL NUMBER OF COMPONENTS. THE EIGENVALUES OF THE COMPONENTS ARE CAPTURED.....  | 37 |
| GRAPH 10 - ACCURACY OF MODELS IN ATTRIBUTE REMOVAL AS A FUNCTION OF THEIR MEMBERSHIP IN PCA COMPONENTS FOR SHAP.....             | 38 |
| GRAPH 11 - SUMMARY PLOT FOR THE IMPACT OF EACH PCA COMPONENT ON THE MODEL OUTPUT FOR THE SHAP METHOD. ....                       | 40 |
| GRAPH 12 - VISUALISATION OF THE IMPORTANCE OF INDIVIDUAL PCA COMPONENTS ACCORDING TO THE PERMUTATION METHOD.....                 | 41 |
| GRAPH 13 - MODEL ACCURACIES FOR SEQUENTIAL COMPONENT REMOVAL USING THE SHAP METHOD. ....                                         | 42 |
| GRAPH 14 - MODEL ACCURACIES FOR SEQUENTIAL COMPONENT REMOVAL USING THE PERMUTATION METHOD.....                                   | 42 |
| GRAPH 15 - AVERAGE VALUES OF ACCURACIES FOR RANDOM INTERCHANGE OF VALUES OUTSIDE THE INTERVAL ACCORDING TO THE ARA METHOD.....   | 45 |

# List of images

|                                                                                                                                              |    |
|----------------------------------------------------------------------------------------------------------------------------------------------|----|
| IMAGE 1 - MODEL WITH SET PARAMETERS USED IN INITIAL MODEL TRAINING AS WELL AS IN TRAINING ON<br>MODIFIED DATASETS FOR LATER COMPARISON. .... | 20 |
| IMAGE 2 - THIS IS THE ACCURACY OF THE INITIAL MODEL. THIS ACCURACY WAS LATER USED AS A<br>BENCHMARK .....                                    | 21 |
| IMAGE 3 - PERFORMANCE OF KMO AND BARTLETT'S TEST AND THEIR RESULTS. ....                                                                     | 32 |
| IMAGE 4 - REMOVAL OF STRONGLY CORRELATED VALUES. ....                                                                                        | 33 |
| IMAGE 5 - PERFORMANCE OF KMO AND BARTLETT'S TEST AND THEIR RESULTS AFTER PROCEDURES WHICH<br>REDUCED CORRELATION OF ATTRIBUTES. ....         | 34 |
| IMAGE 6 - PERFORMING PCA BEFORE FACTOR ANALYSIS. ....                                                                                        | 35 |
| IMAGE 7 - PERFORM FACTOR ANALYSIS ON THE PCA OUTPUT WITH THE NUMBER OF FACTORS SET TO 20. .                                                  | 35 |
| IMAGE 8 - CODE AND RESULTS FOR THE ARA METHOD.....                                                                                           | 44 |

# List of tables

TABLE 1 - RESULTS OF THE SHAP/ARA SEQUENTIAL ATTRIBUTE REMOVAL PROCEDURE (OWN PROCESSING) .....26

TABLE 2 - MEASUREMENT RESULTS FOR THE ARA/SHAP STEPWISE ADDITION METHOD (OWN PROCESSING) .....30

TABLE 3 - THE IMPORTANCE OF EACH COMPONENT ACCORDING TO THE 3 METHODS USED (OWN PROCESSING) .....47

TABLE 4 - MODEL ACCURACIES FOR SELECTED METHODS (OWN PROCESSING) .....47

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